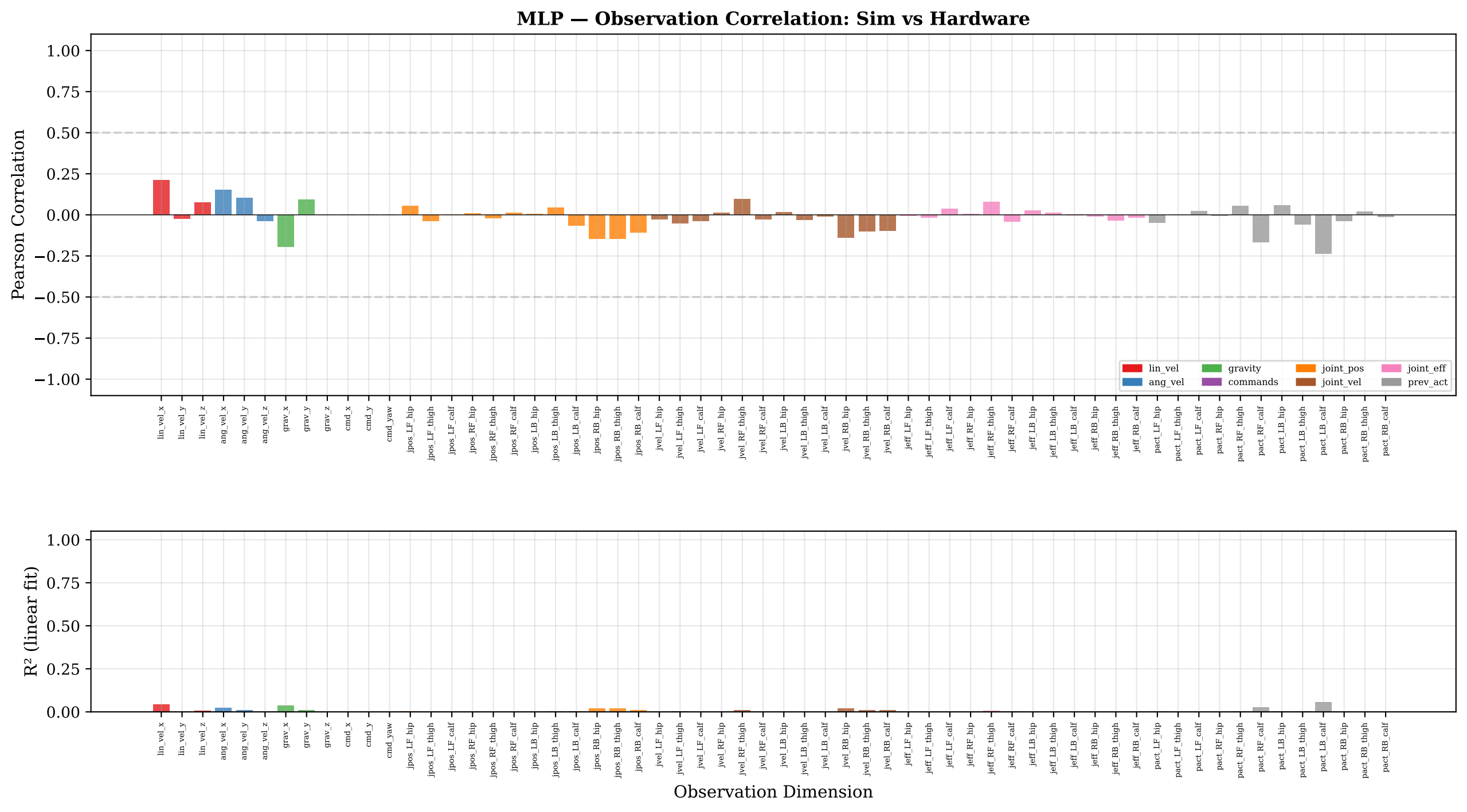
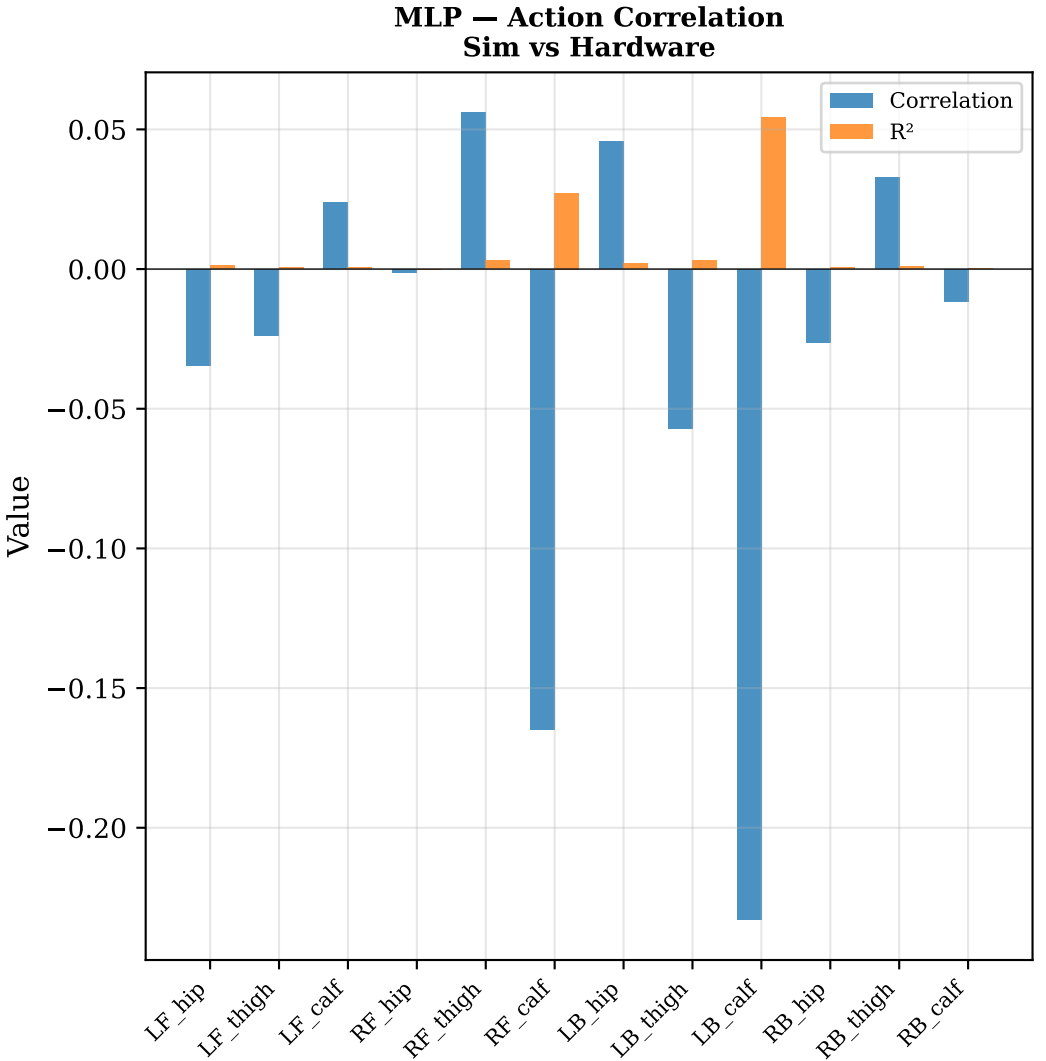
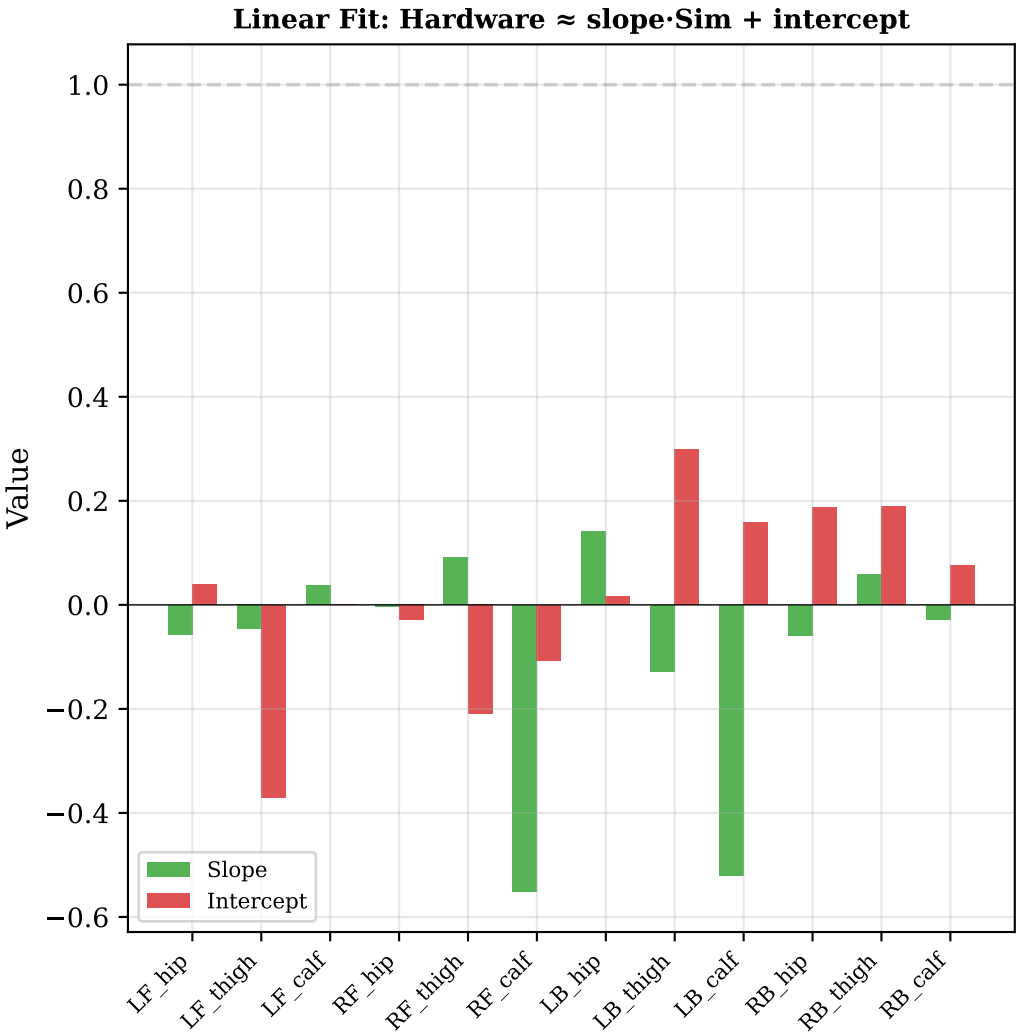
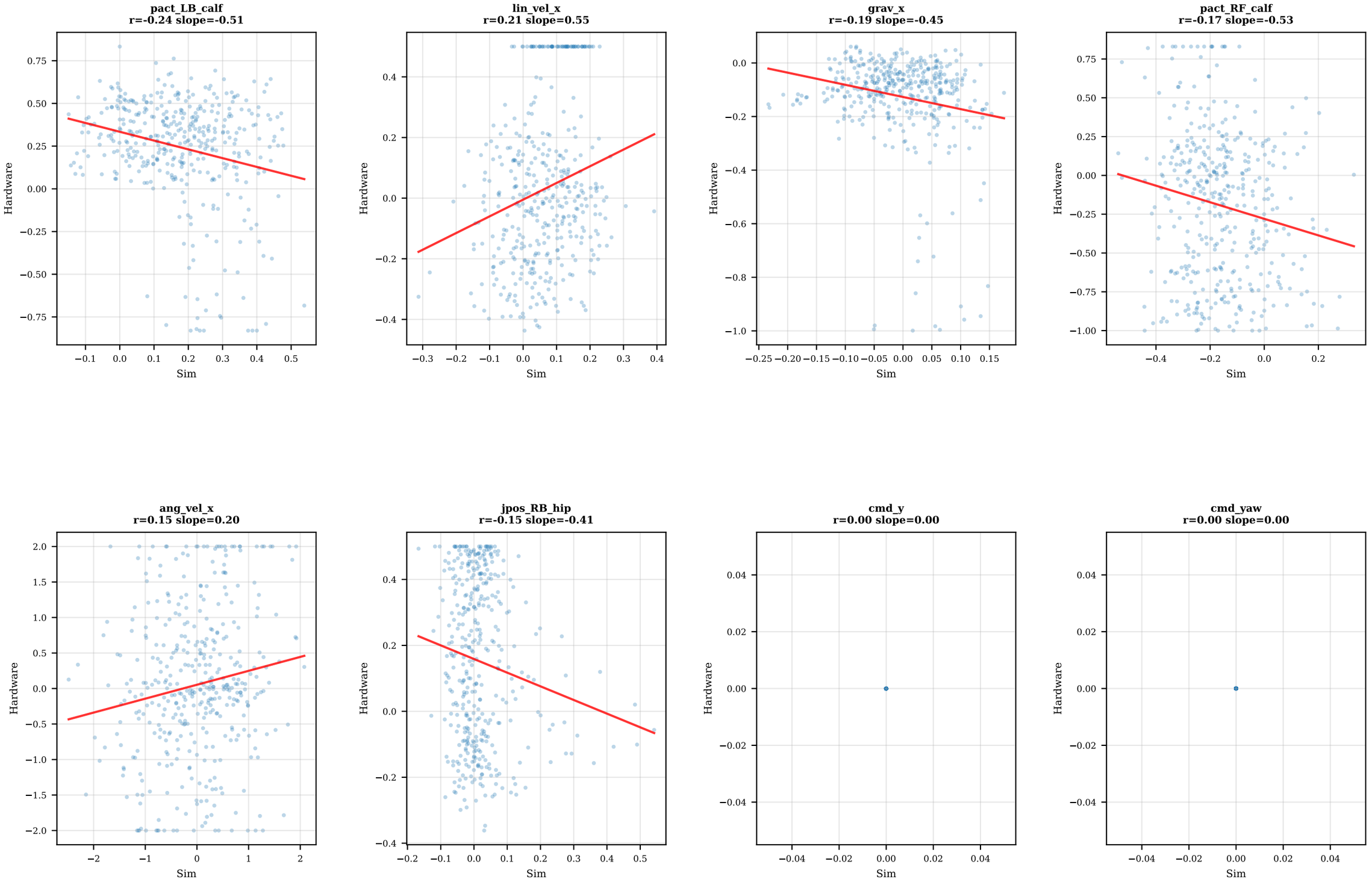

MLP — Observation Correlation: Sim vs Hardware
Pearson Correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
lin_vel
ang_vel
gravity
commands
joint_pos
joint_vel
joint_eff
prev_act
lin_vel_x
lin_vel_y
lin_vel_z
ang_vel_x
ang_vel_y
ang_vel_z
grav_x
grav_y
grav_z
cmd_x
cmd_y
cmd_yaw
jpos_LF_hip
jpos_LF_thigh
jpos_LF_calf
jpos_RF_hip
jpos_RF_thigh
jpos_RF_calf
jpos_LB_hip
jpos_LB_thigh
jpos_LB_calf
jpos_RB_hip
jpos_RB_thigh
jpos_RB_calf
jvel_LF_hip
jvel_LF_thigh
jvel_LF_calf
jvel_RF_hip
jvel_RF_thigh
jvel_RF_calf
jvel_LB_hip
jvel_LB_thigh
jvel_LB_calf
jvel_RB_hip
jvel_RB_thigh
jvel_RB_calf
jeff_LF_hip
jeff_LF_thigh
jeff_LF_calf
jeff_RF_hip
jeff_RF_thigh
jeff_RF_calf
jeff_LB_hip
jeff_LB_thigh
jeff_LB_calf
jeff_RB_hip
jeff_RB_thigh
jeff_RB_calf
pact_LF_hip
pact_LF_thigh
pact_LF_calf
pact_RF_hip
pact_RF_thigh
pact_RF_calf
pact_LB_hip
pact_LB_thigh
pact_LB_calf
pact_RB_hip
pact_RB_thigh
pact_RB_calf
R² (linear fit)
1.00
0.75
0.50
0.25
0.00
Observation Dimension

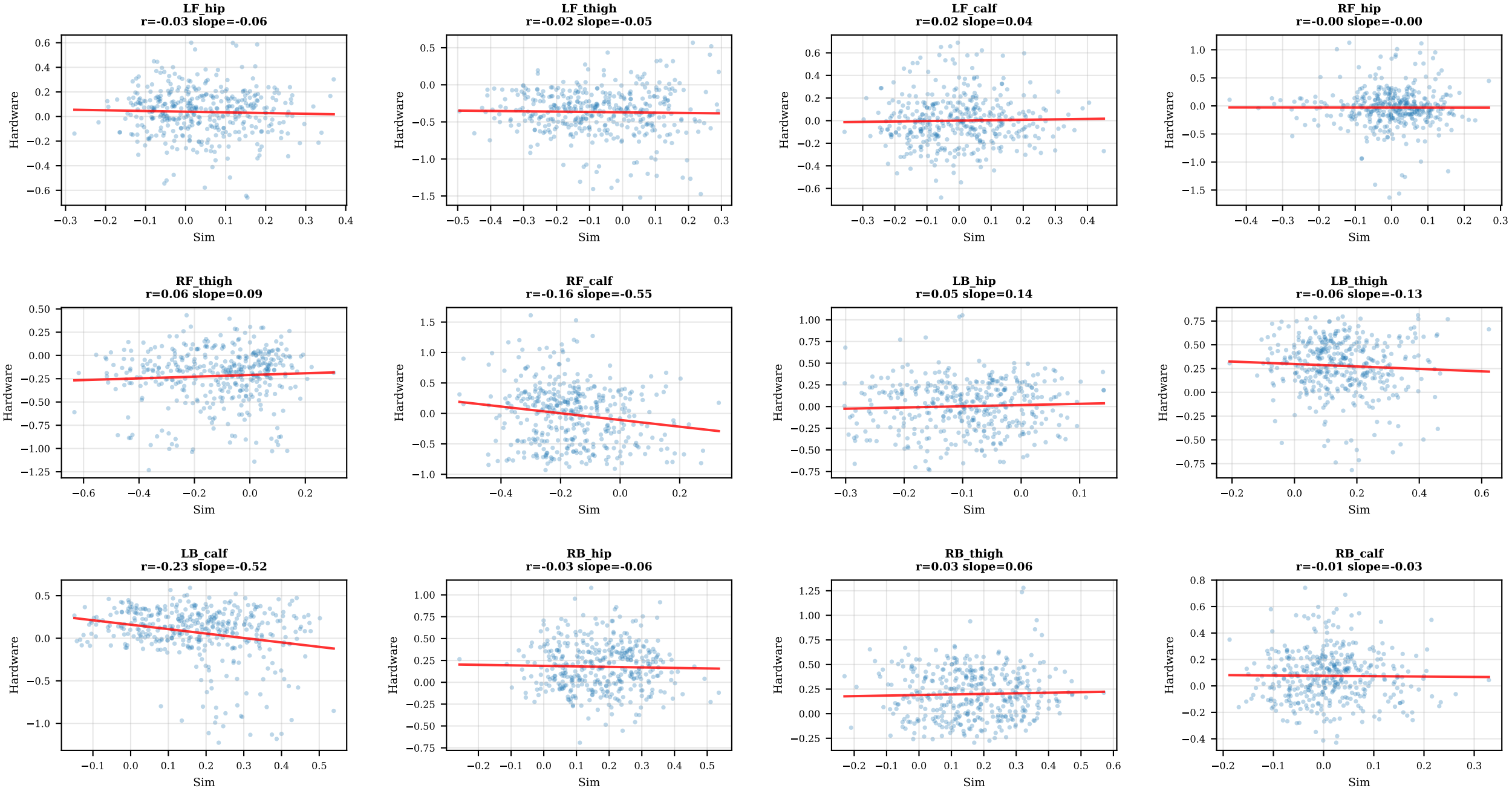

MLP — Action Correlation
Sim vs Hardware
Correlation
R²
Value
0.05
0.00
−0.05
−0.10
−0.15
−0.20
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf

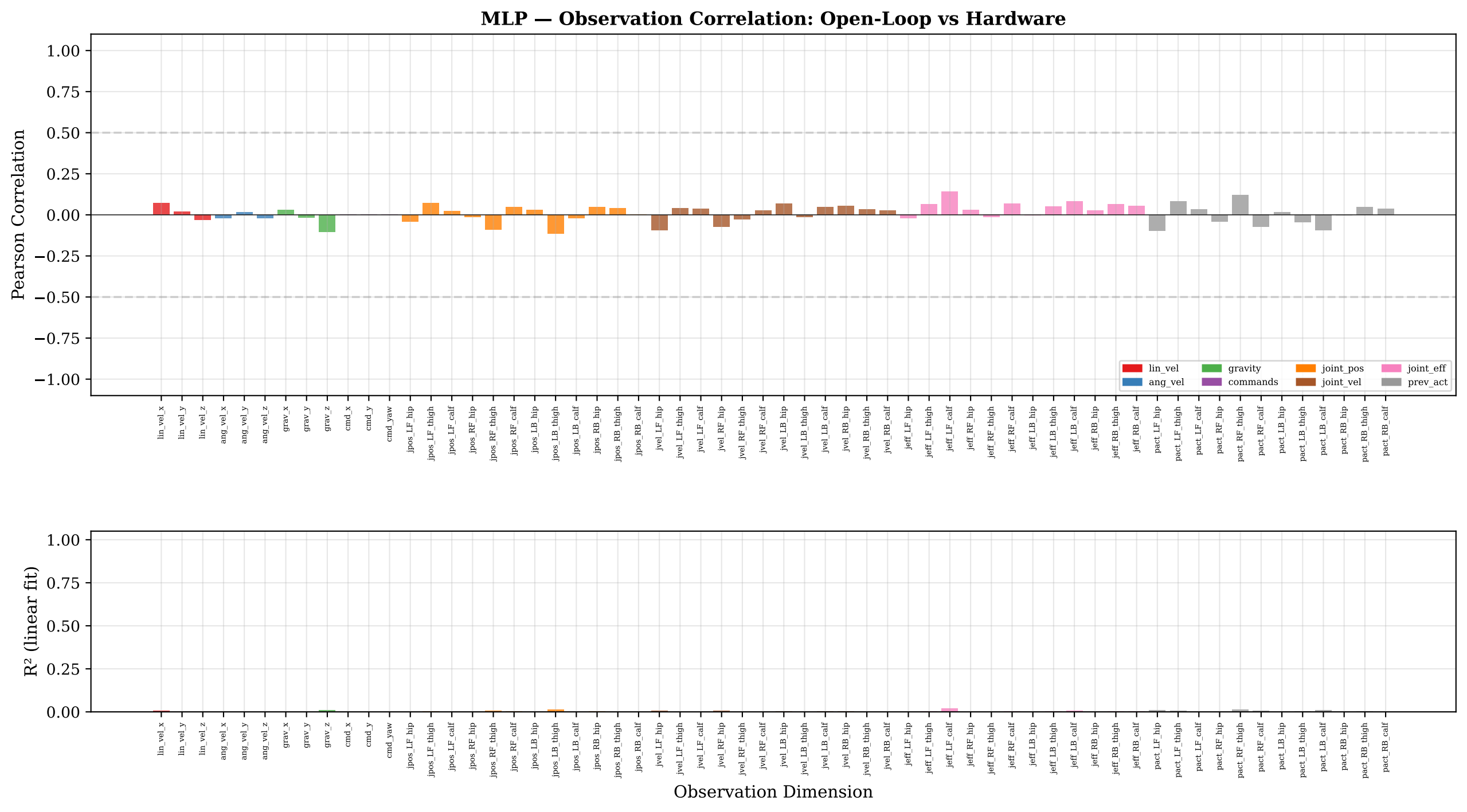

Linear Fit: Hardware ≈ slope·Sim + intercept
Value
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Slope
Intercept
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf

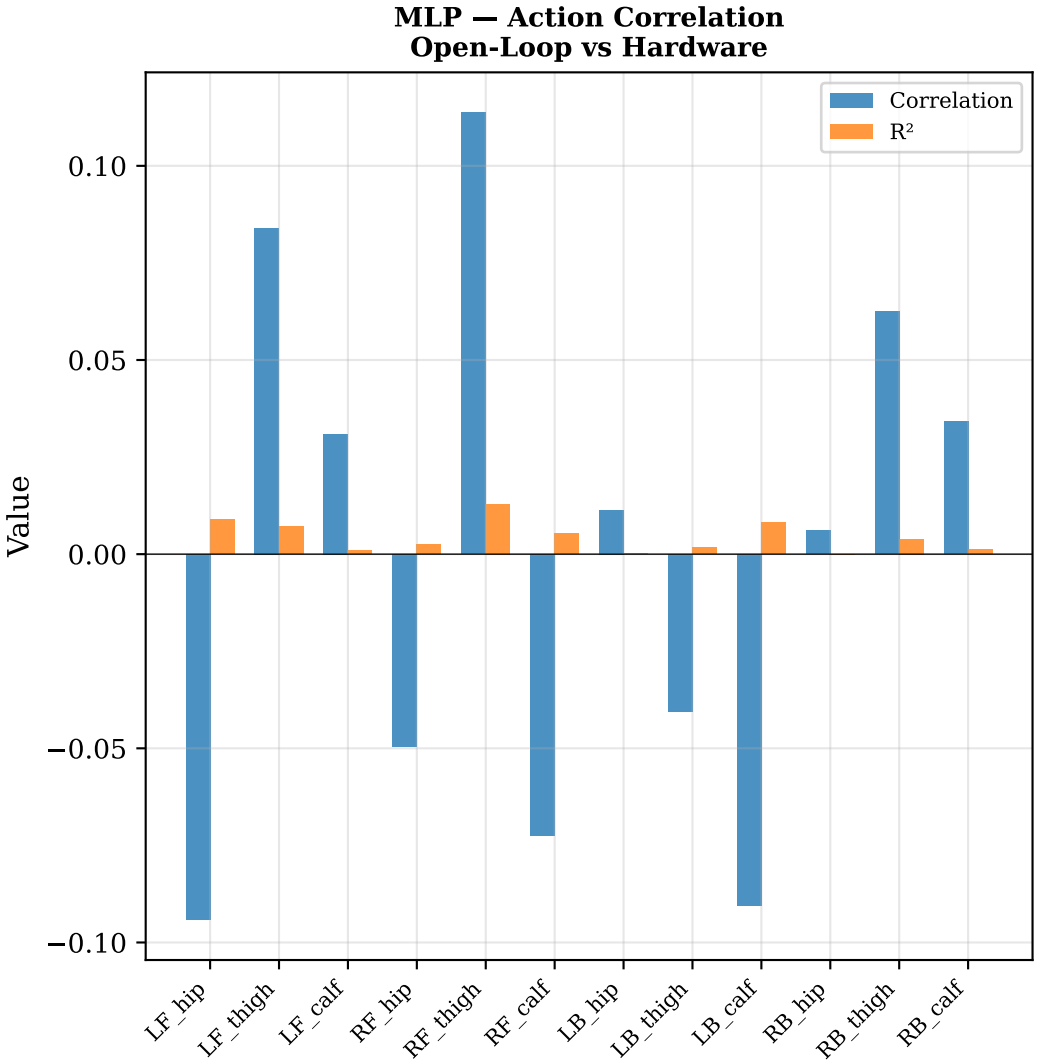
MLP — Key Dimensions: Sim vs Hardware

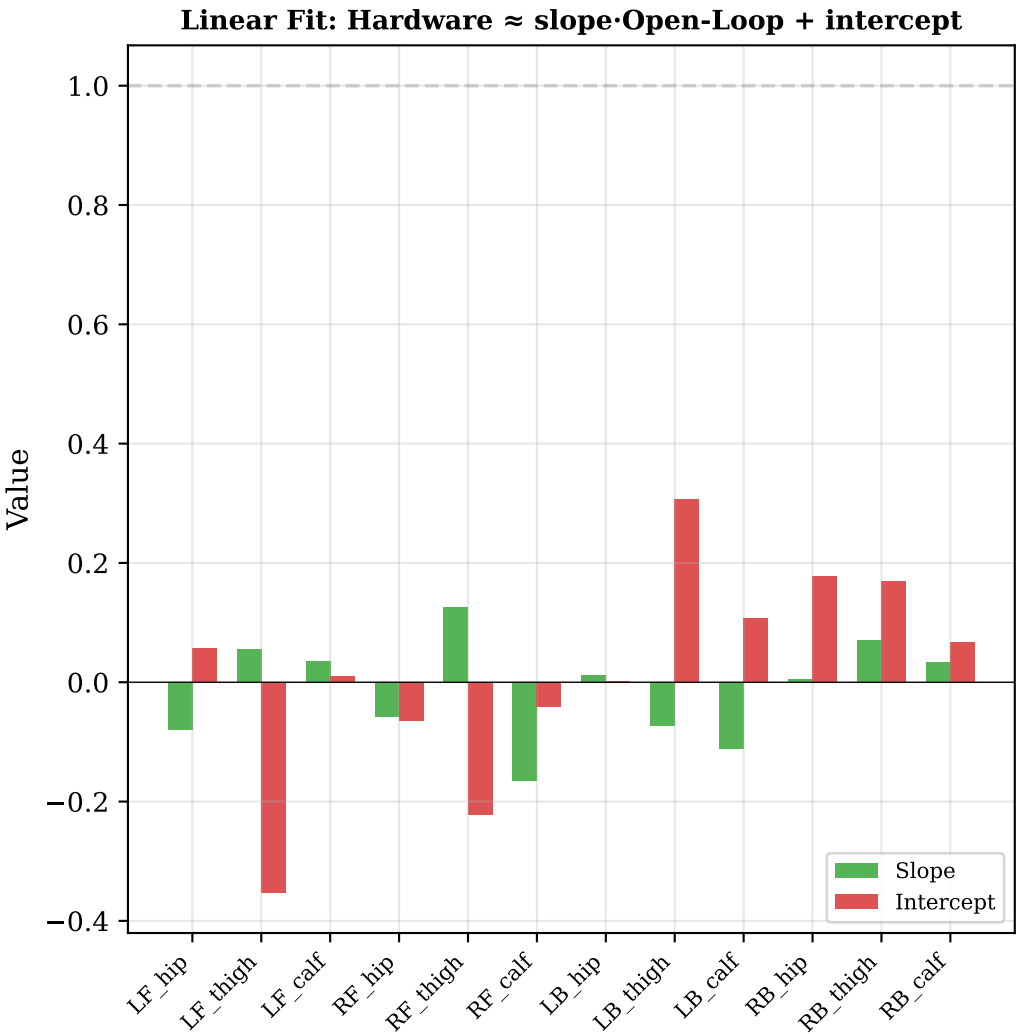


MLP — Key Dimensions: Sim vs Hardware

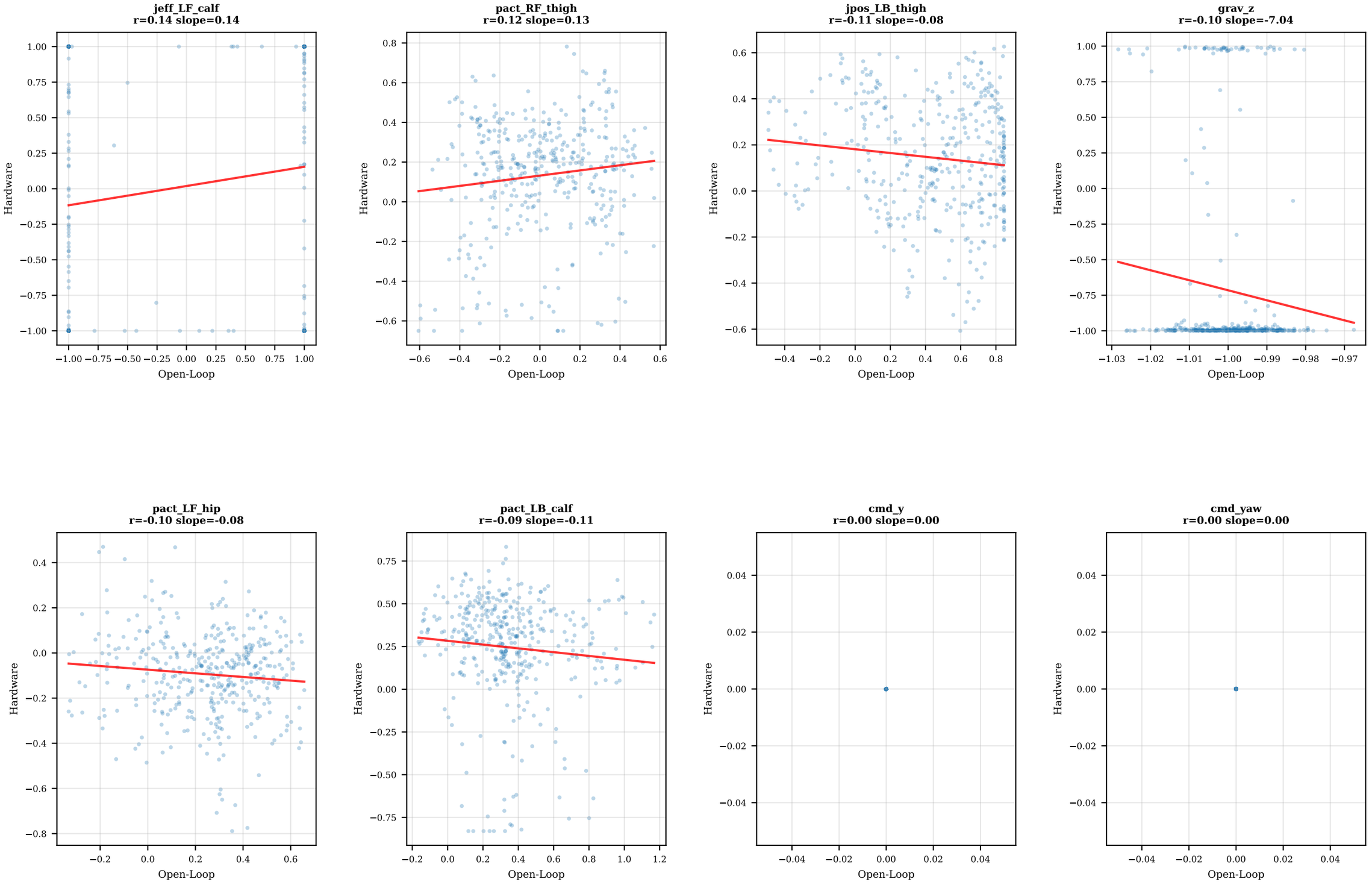



MLP — Observation Correlation: Open-Loop vs Hardware
Pearson Correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
lin_vel
ang_vel
gravity
commands
joint_pos
joint_vel
joint_eff
prev_act
lin_vel_x
lin_vel_y
lin_vel_z
ang_vel_x
ang_vel_y
ang_vel_z
grav_x
grav_y
grav_z
cmd_x
cmd_y
cmd_yaw
jpos_LF_hip
jpos_LF_thigh
jpos_LF_calf
jpos_RF_hip
jpos_RF_thigh
jpos_RF_calf
jpos_LB_hip
jpos_LB_thigh
jpos_LB_calf
jpos_RB_hip
jpos_RB_thigh
jpos_RB_calf
jvel_LF_hip
jvel_LF_thigh
jvel_LF_calf
jvel_RF_hip
jvel_RF_thigh
jvel_RF_calf
jvel_LB_hip
jvel_LB_thigh
jvel_LB_calf
jvel_RB_hip
jvel_RB_thigh
jvel_RB_calf
jeff_LF_hip
jeff_LF_thigh
jeff_LF_calf
jeff_RF_hip
jeff_RF_thigh
jeff_RF_calf
jeff_LB_hip
jeff_LB_thigh
jeff_LB_calf
jeff_RB_hip
jeff_RB_thigh
jeff_RB_calf
pact_LF_hip
pact_LF_thigh
pact_LF_calf
pact_RF_hip
pact_RF_thigh
pact_RF_calf
pact_LB_hip
pact_LB_thigh
pact_LB_calf
pact_RB_hip
pact_RB_thigh
pact_RB_calf
R² (linear fit)
1.00
0.75
0.50
0.25
0.00
Observation Dimension

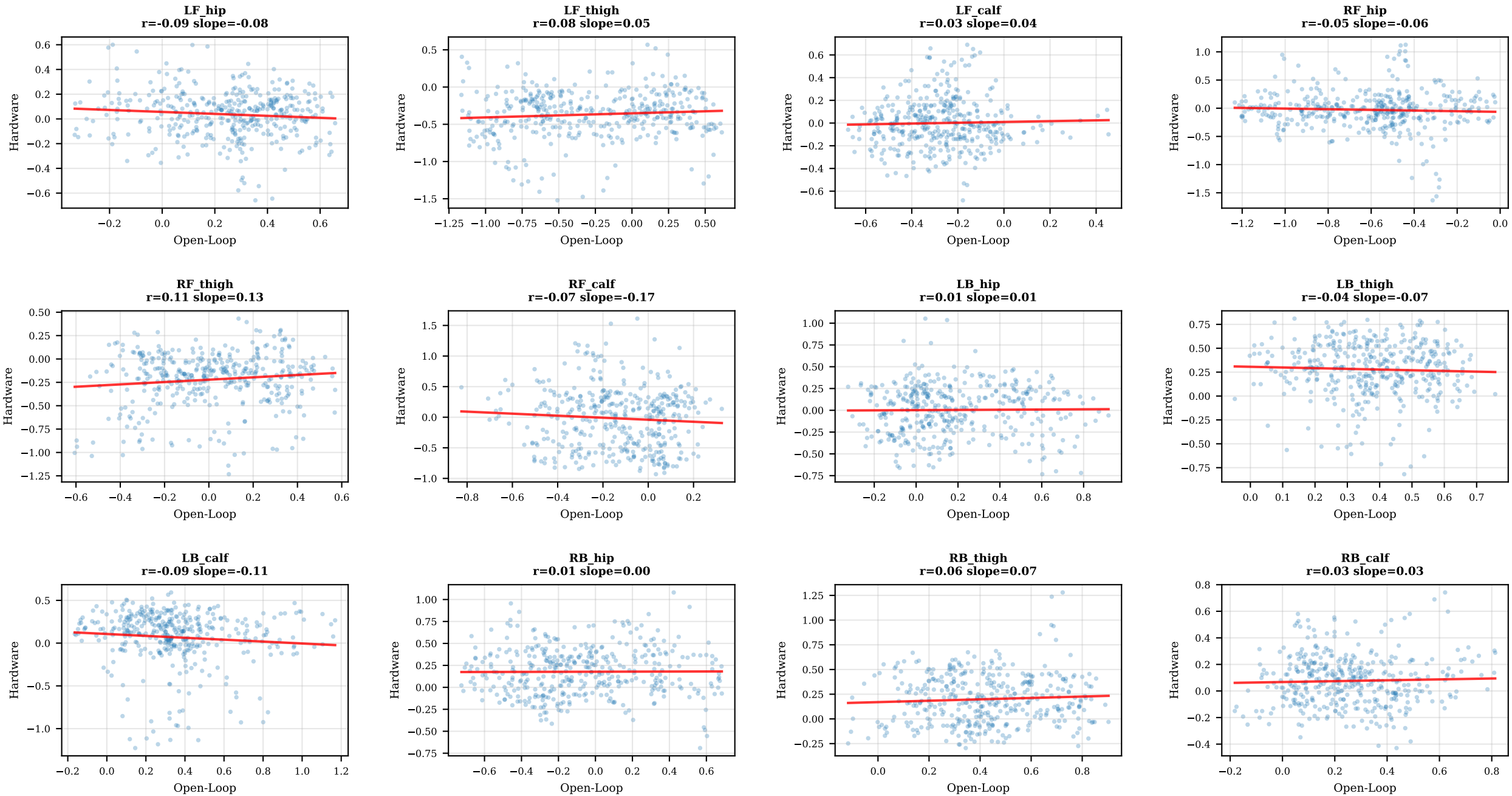

MLP — Action Correlation
Open-Loop vs Hardware
Correlation
R²
Value
0.10
0.05
0.00
−0.05
−0.10
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf

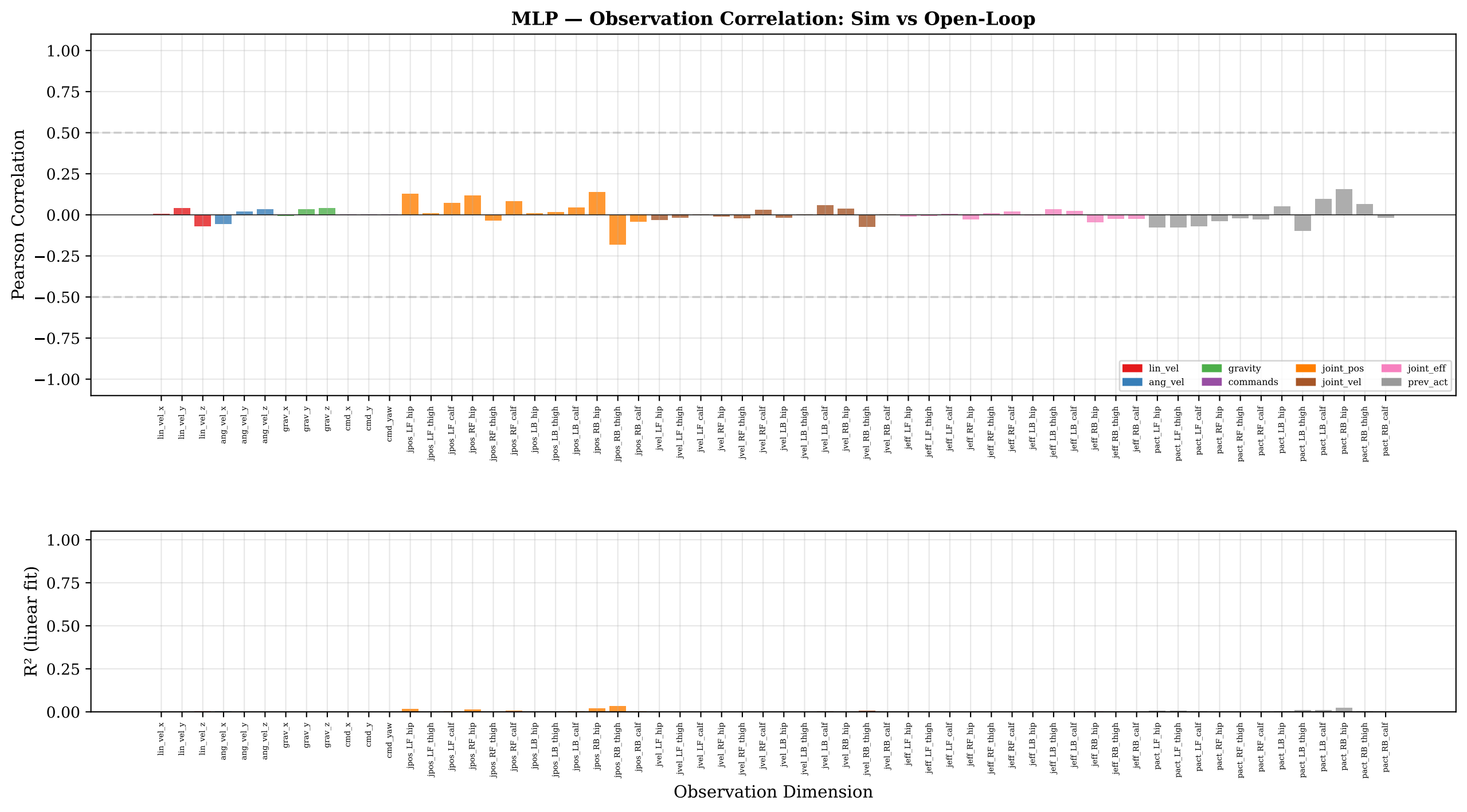

Linear Fit: Hardware ≈ slope·Open-Loop + intercept
Value
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
Slope
Intercept
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf

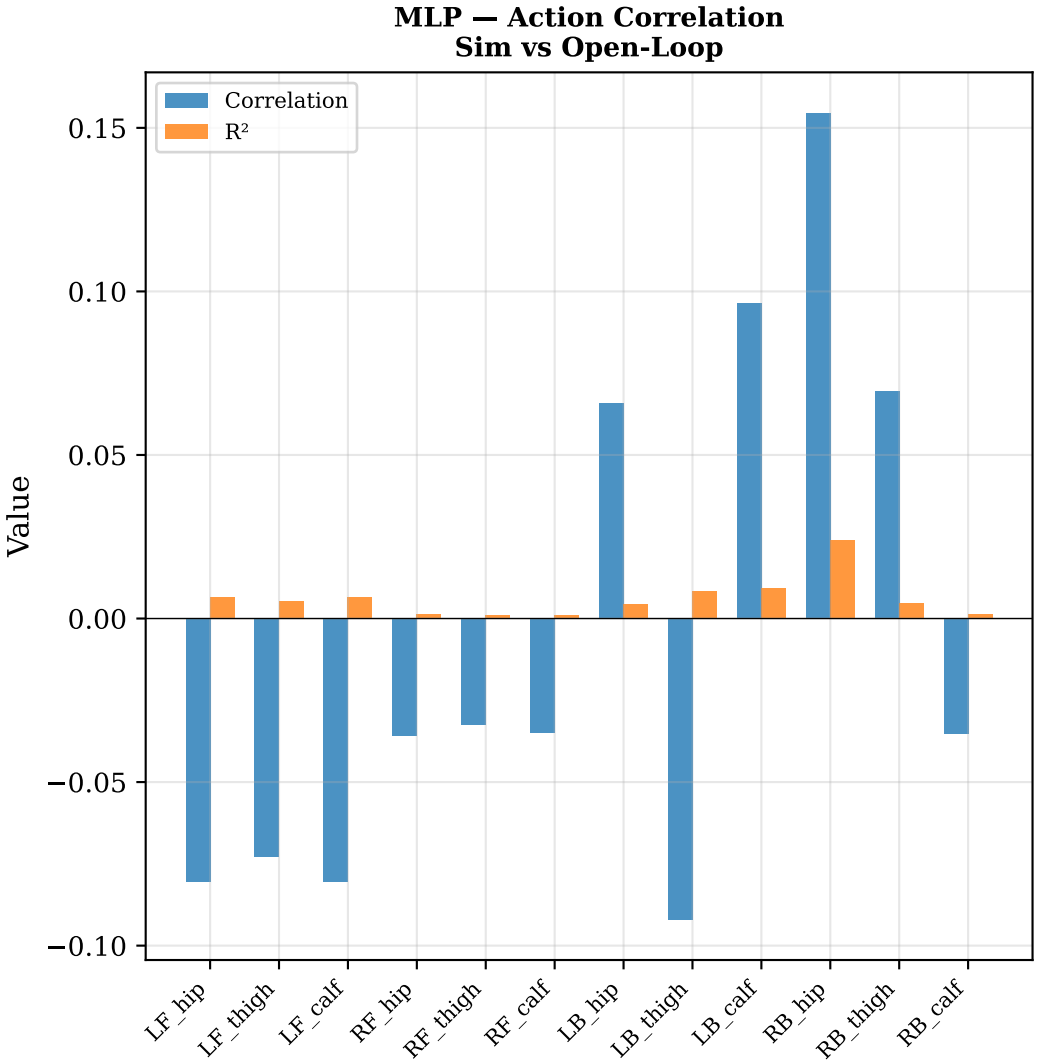
MLP — Key Dimensions: Open-Loop vs Hardware

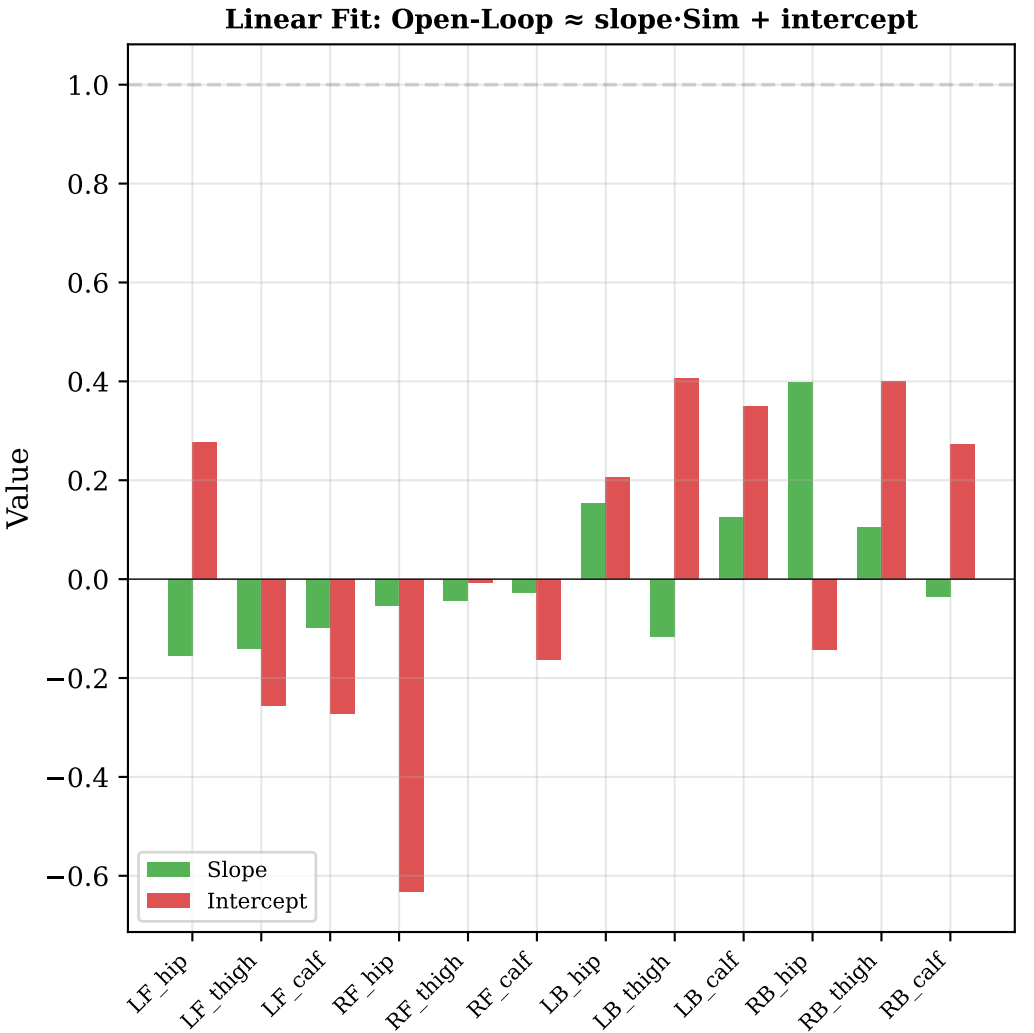


MLP — Key Dimensions: Open-Loop vs Hardware

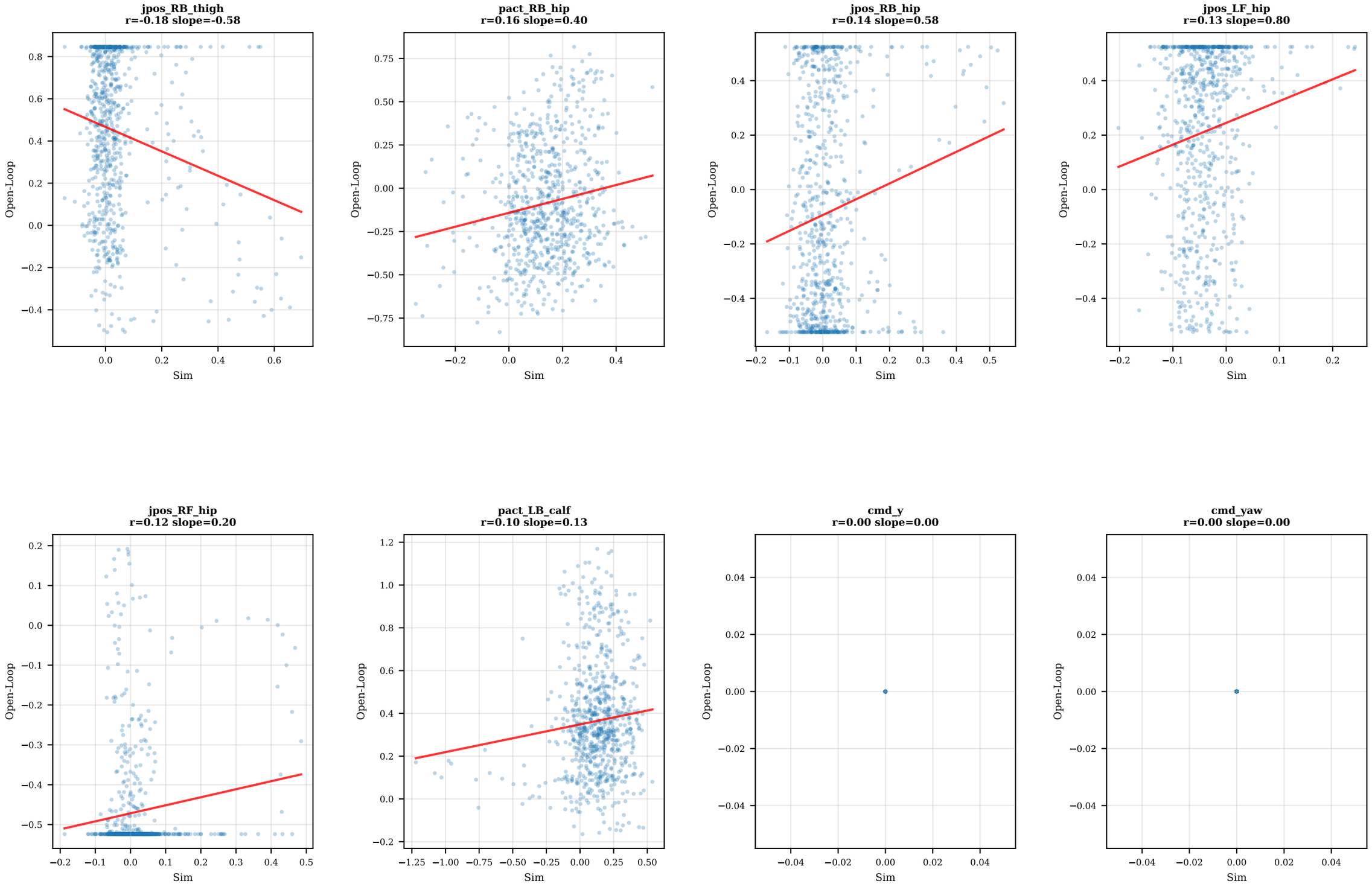



MLP — Observation Correlation: Sim vs Open-Loop
Pearson Correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
lin_vel
ang_vel
gravity
commands
joint_pos
joint_vel
joint_eff
prev_act
lin_vel_x
lin_vel_y
lin_vel_z
ang_vel_x
ang_vel_y
ang_vel_z
grav_x
grav_y
grav_z
cmd_x
cmd_y
cmd_yaw
jpos_LF_hip
jpos_LF_thigh
jpos_LF_calf
jpos_RF_hip
jpos_RF_thigh
jpos_RF_calf
jpos_LB_hip
jpos_LB_thigh
jpos_LB_calf
jpos_RB_hip
jpos_RB_thigh
jpos_RB_calf
jvel_LF_hip
jvel_LF_thigh
jvel_LF_calf
jvel_RF_hip
jvel_RF_thigh
jvel_RF_calf
jvel_LB_hip
jvel_LB_thigh
jvel_LB_calf
jvel_RB_hip
jvel_RB_thigh
jvel_RB_calf
jeff_LF_hip
jeff_LF_thigh
jeff_LF_calf
jeff_RF_hip
jeff_RF_thigh
jeff_RF_calf
jeff_LB_hip
jeff_LB_thigh
jeff_LB_calf
jeff_RB_hip
jeff_RB_thigh
jeff_RB_calf
pact_LF_hip
pact_LF_thigh
pact_LF_calf
pact_RF_hip
pact_RF_thigh
pact_RF_calf
pact_LB_hip
pact_LB_thigh
pact_LB_calf
pact_RB_hip
pact_RB_thigh
pact_RB_calf
R² (linear fit)
1.00
0.75
0.50
0.25
0.00
Observation Dimension

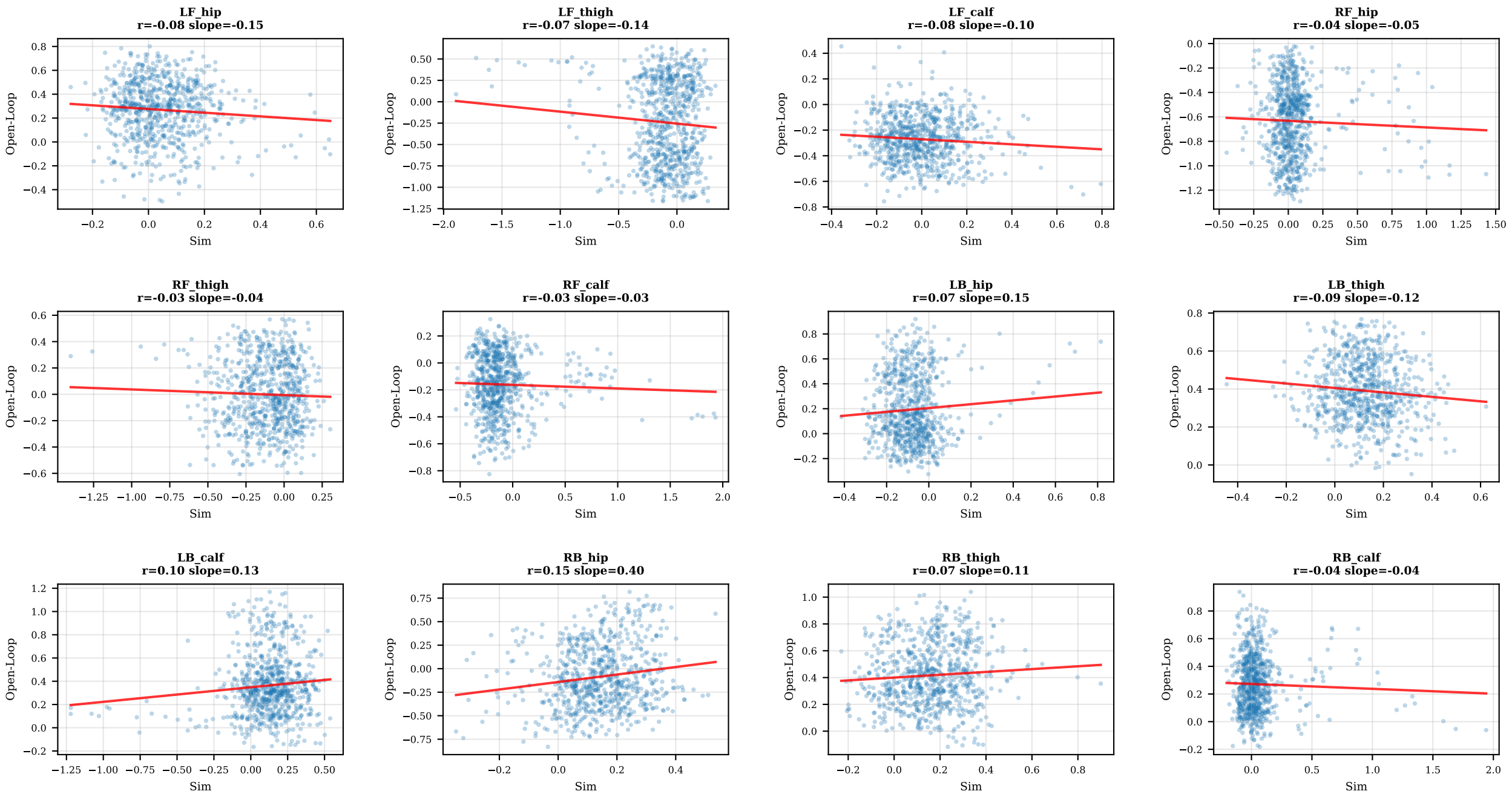

MLP — Action Correlation
Sim vs Open-Loop
Correlation
R²
Value
0.15
0.10
0.05
0.00
−0.05
−0.10
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf


Linear Fit: Open-Loop ≈ slope·Sim + intercept
Value
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Slope
Intercept
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf

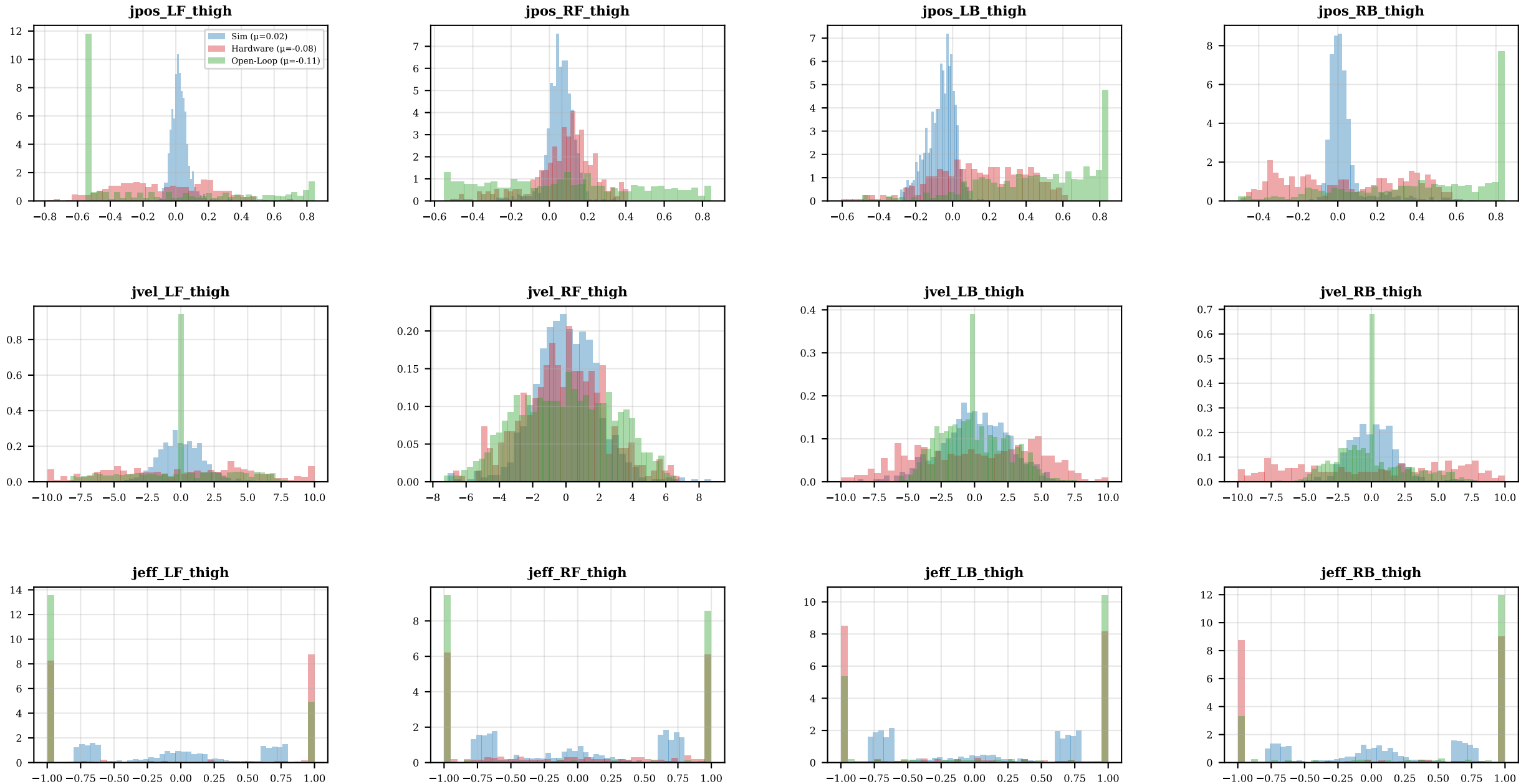
MLP — Key Dimensions: Sim vs Open-Loop



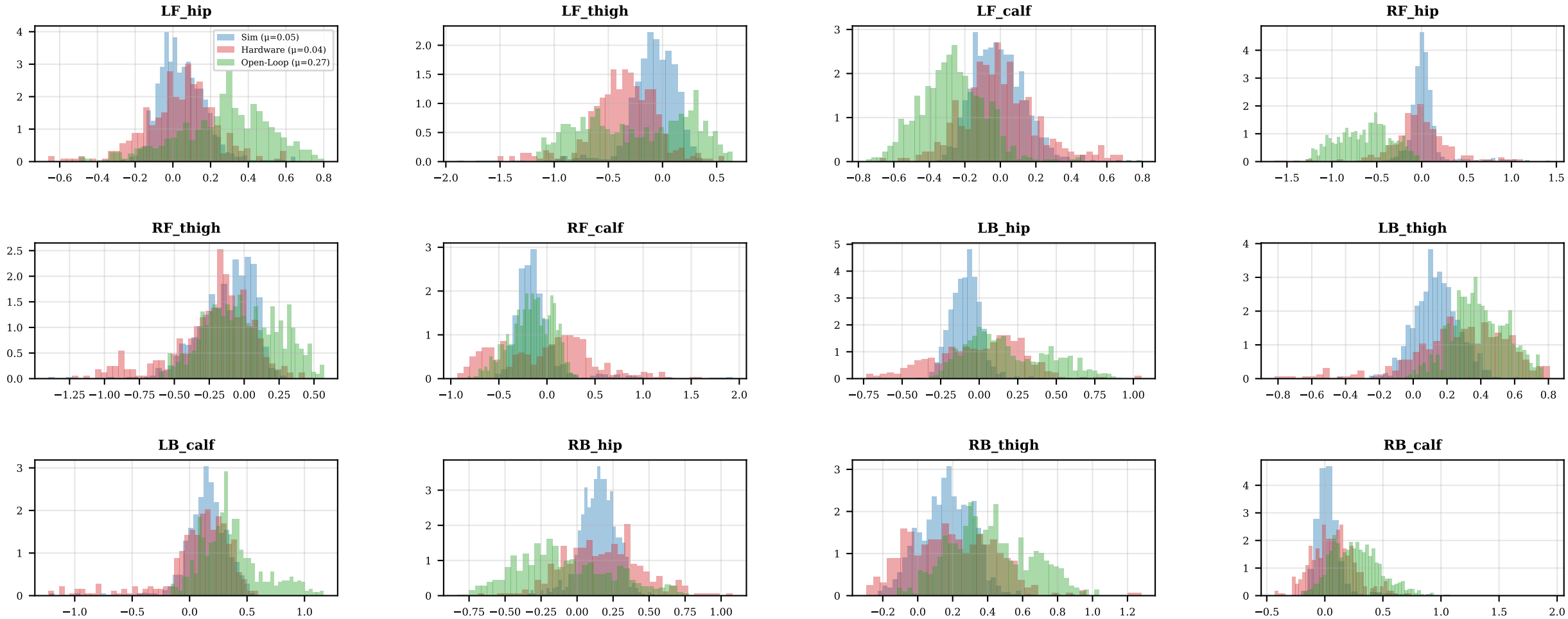
MLP — Key Dimensions: Sim vs Open-Loop



MLP — Thigh Obs Distributions Across Domains



MLP — Action Distributions Across Domains



MLP — Full Dimension-by-Dimension Linear Relationships

MLP — LINEAR RELATIONSHIPS: Sim/Open-Loop → Hardware

Sim → Hardware:

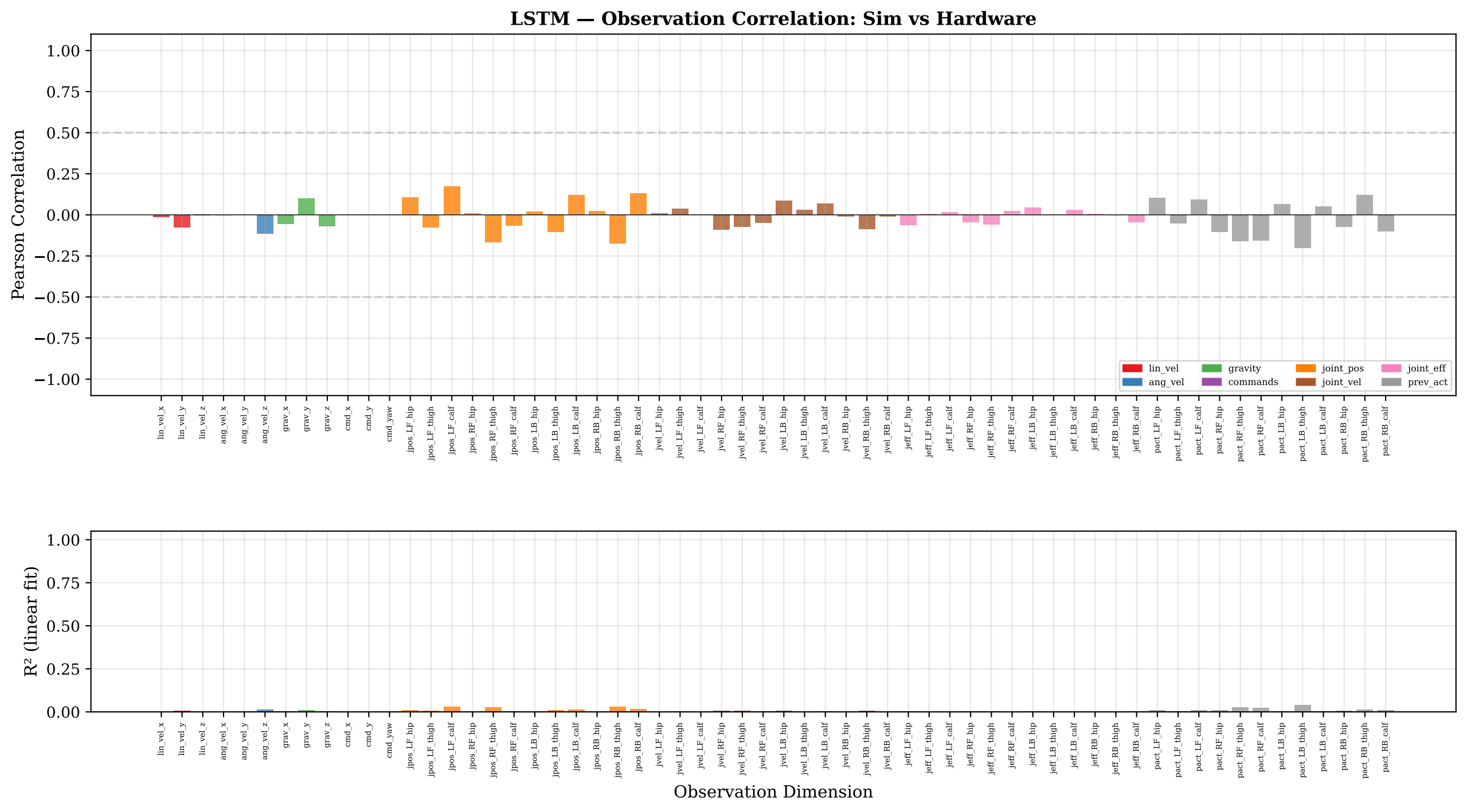
Dimension	Corr	Slope	Intcpt	R^2	μ_src	μ_tgt	σ_src	σ_tgt
OBSERVATIONS:								
lin_vel_x	+0.212	0.551	-0.005	0.045	0.060	0.028	0.099	0.258
lin_vel_y	-0.025	-0.039	0.244	0.001	0.014	0.243	0.088	0.136
lin_vel_z	+0.074	0.106	0.131	0.005	-0.003	0.131	0.091	0.130
ang_vel_x	+0.151	0.196	0.053	0.023	-0.018	0.049	0.810	1.051
ang_vel_y	+0.103	0.135	-0.176	0.011	0.016	-0.174	0.879	1.149
ang_vel_z	-0.036	-0.073	0.248	0.001	0.159	0.237	0.559	1.136
grav_x	-0.194	-0.454	-0.127	0.038	-0.004	-0.125	0.073	0.171
grav_y	+0.094	0.058	0.032	0.009	-0.030	0.030	0.076	0.047
grav_z	+0.003	0.029	-0.688	0.000	-0.998	-0.718	0.058	0.661
cmd_x	+0.000	0.000	0.000	0.000	0.094	0.150	0.073	0.000
cmd_y	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
cmd_yaw	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
jpos_LF_hip	+0.053	0.201	-0.037	0.003	-0.043	-0.046	0.044	0.165
jpos_LF_thigh	-0.037	-0.237	-0.079	0.001	0.022	-0.084	0.045	0.287
jpos_LF_calf	+0.003	0.012	0.066	0.000	0.083	0.067	0.062	0.230
jpos_RF_hip	+0.008	0.038	0.252	0.000	-0.002	0.252	0.043	0.193
jpos_RF_thigh	-0.019	-0.046	0.076	0.000	0.063	0.073	0.071	0.171
jpos_RF_calf	+0.011	0.089	-0.086	0.000	-0.037	-0.089	0.065	0.532
jpos_LB_hip	+0.006	0.023	-0.034	0.000	0.070	-0.033	0.075	0.286
jpos_LB_thigh	+0.045	0.155	0.155	0.002	-0.062	0.145	0.072	0.250
jpos_LB_calf	-0.066	-0.215	0.144	0.004	0.100	0.123	0.084	0.274
jpos_RB_hip	-0.146	-0.414	0.159	0.021	0.018	0.151	0.087	0.246
jpos_RB_thigh	-0.144	-0.466	0.058	0.021	0.020	0.049	0.093	0.301
jpos_RB_calf	-0.106	-0.176	-0.065	0.011	-0.066	-0.054	0.107	0.179
jvel_LF_hip	-0.027	-0.053	0.063	0.001	-0.074	0.067	1.641	3.143
jvel_LF_thigh	-0.050	-0.158	0.097	0.002	-0.036	0.102	1.674	5.308
jvel_LF_calf	-0.036	-0.096	0.056	0.001	0.023	0.054	1.537	4.134
jvel_RF_hip	+0.012	0.028	-0.060	0.000	-0.023	-0.061	1.456	3.447
jvel_RF_thigh	+0.095	0.127	-0.100	0.009	0.002	-0.099	1.876	2.510
jvel_RF_calf	-0.028	-0.111	0.045	0.001	-0.022	0.048	2.054	8.155
jvel_LB_hip	+0.017	0.042	0.014	0.000	0.016	0.015	2.182	5.417
jvel_LB_thigh	-0.030	-0.051	0.027	0.001	-0.108	0.032	2.614	4.455
jvel_LB_calf	-0.010	-0.026	-0.117	0.000	0.073	-0.119	1.894	4.610
jvel_RB_hip	-0.139	-0.289	-0.115	0.019	-0.063	-0.096	2.296	4.758
jvel_RB_thigh	-0.101	-0.301	-0.005	0.010	0.001	-0.005	1.934	5.733
jvel_RB_calf	-0.096	-0.141	-0.036	0.009	-0.039	-0.030	2.277	3.350
jeff_LF_hip	-0.006	-0.010	0.012	0.000	-0.018	0.012	0.583	0.892
jeff_LF_thigh	-0.018	-0.030	0.041	0.000	-0.000	0.041	0.571	0.947
jeff_LF_calf	+0.036	0.062	0.005	0.001	0.018	0.006	0.546	0.939
jeff_RF_hip	+0.006	0.009	0.020	0.000	-0.003	0.020	0.531	0.864
jeff_RF_thigh	+0.079	0.123	0.002	0.006	-0.036	-0.002	0.548	0.852
jeff_RF_calf	-0.042	-0.069	-0.059	0.002	0.010	-0.060	0.593	0.977
jeff_LB_hip	+0.026	0.041	-0.063	0.001	0.002	-0.063	0.610	0.944
jeff_LB_thigh	+0.012	0.017	-0.027	0.000	-0.015	-0.028	0.631	0.933
jeff_LB_calf	-0.004	-0.007	0.033	0.000	0.049	0.033	0.532	0.944
jeff_RB_hip	-0.008	-0.012	0.020	0.000	0.001	0.020	0.617	0.939
jeff_RB_thigh	-0.034	-0.060	0.023	0.001	0.052	0.020	0.545	0.958
jeff_RB_calf	-0.017	-0.026	0.003	0.000	0.009	0.003	0.609	0.920
pact_LF_hip	-0.047	-0.077	-0.091	0.002	0.041	-0.094	0.114	0.186
pact_LF_thigh	+0.002	0.003	-0.175	0.000	-0.074	-0.175	0.159	0.286
pact_LF_calf	+0.024	0.039	0.129	0.001	-0.003	0.129	0.135	0.216
pact_RF_hip	-0.007	-0.018	0.529	0.000	-0.000	0.529	0.099	0.273
pact_RF_thigh	+0.055	0.088	0.143	0.003	-0.114	0.133	0.177	0.283
pact_RF_calf	-0.166	-0.534	-0.280	0.028	-0.159	-0.195	0.147	0.472
pact_LB_hip	+0.057	0.174	-0.062	0.003	-0.093	-0.078	0.092	0.282
pact_LB_thigh	-0.059	-0.134	0.310	0.004	0.149	0.290	0.126	0.284
pact_LB_calf	-0.237	-0.515	0.334	0.056	0.175	0.244	0.143	0.310
pact_RB_hip	-0.037	-0.082	0.369	0.001	0.171	0.355	0.119	0.264
pact_RB_thigh	+0.020	0.034	0.091	0.000	0.164	0.096	0.140	0.239
pact_RB_calf	-0.012	-0.028	-0.104	0.000	0.012	-0.105	0.080	0.189
ACTIONS:								
LF_hip	-0.035	-0.057	0.039	0.001	0.041	0.037	0.114	0.188
LF_thigh	-0.024	-0.047	-0.371	0.001	-0.074	-0.368	0.160	0.314
LF_calf	+0.024	0.038	0.000	0.001	-0.002	0.000	0.136	0.215
RF_hip	-0.001	-0.004	-0.029	0.000	0.000	-0.029	0.099	0.344
RF_thigh	+0.056	0.091	-0.210	0.003	-0.115	-0.220	0.178	0.289
RF_calf	-0.165	-0.551	-0.107	0.027	-0.161	-0.019	0.146	0.489
LB_hip	+0.046	0.141	0.017	0.002	-0.094	0.004	0.092	0.285
LB_thigh	-0.057	-0.129	0.298	0.003	0.150	0.279	0.126	0.284
LB_calf	-0.233	-0.521	0.159	0.054	0.177	0.068	0.144	0.321
RB_hip	-0.026	-0.059	0.188	0.001	0.173	0.178	0.119	0.269
RB_thigh	+0.033	0.058	0.190	0.001	0.165	0.200	0.140	0.247
RB_calf	-0.012	-0.028	0.076	0.000	0.012	0.076	0.081	0.189

Open-Loop → Hardware:

Dimension	Corr	Slope	Intcpt	R^2	μ_src	μ_tgt	σ_src	σ_tgt
OBSERVATIONS:								
lin_vel_x	+0.072	0.886	0.029	0.005	-0.001	0.028	0.021	0.258
lin_vel_y	+0.020	0.136	0.243	0.000	-0.000	0.243	0.020	0.136
lin_vel_z	-0.032	-0.206	0.131	0.001	0.000	0.131	0.020	0.130
ang_vel_x	-0.021	-0.218	0.048	0.000	-0.008	0.049	0.100	1.051
ang_vel_y	+0.015	0.167	-0.172	0.000	-0.011	-0.174	0.103	1.149
ang_vel_z	-0.020	-0.227	0.238	0.000	0.006	0.237	0.098	1.136
grav_x	+0.031	0.534	-0.125	0.001	-0.000	-0.125	0.010	0.171
grav_y	-0.018	-0.085	0.030	0.000	-0.000	0.030	0.010	0.047
grav_z	-0.104	-7.036	-7.751	0.011	-1.000	-0.718	0.010	0.661
cmd_x	+0.000	0.000	0.000	0.000	0.150	0.150	0.000	0.000
cmd_y	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
cmd_yaw	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
jpos_LF_hip	-0.039	-0.020	-0.042	0.002	0.215	-0.046	0.327	0.165
jpos_LF_thigh	+0.071	0.042	-0.079	0.005	-0.114	-0.084	0.491	0.287
jpos_LF_calf	+0.022	0.007	0.069	0.000	-0.332	0.067	0.682	0.230
jpos_RF_hip	-0.014	-0.016	0.245	0.000	-0.461	0.252	0.161	0.193
jpos_RF_thigh	-0.090	-0.039	0.076	0.008	0.077	0.073	0.401	0.171
jpos_RF_calf	+0.046	0.037	-0.082	0.002	-0.193	-0.089	0.663	0.532
jpos_LB_hip	+0.029	0.025	-0.037	0.001	0.159	-0.033	0.327	0.286
jpos_LB_thigh	-0.114	-0.082	0.181	0.013	0.441	0.145	0.345	0.250
jpos_LB_calf	-0.020	-0.015	0.129	0.000	0.433	0.123	0.369	0.274
jpos_RB_hip	+0.048	0.029	0.153	0.002	-0.070	0.151	0.403	0.246
jpos_RB_thigh	+0.039	0.034	0.033	0.002	0.465	0.049	0.349	0.301
jpos_RB_calf	+0.003	0.001	-0.054	0.000	0.318	-0.054	0.484	0.179
jvel_LF_hip	-0.092	-0.115	0.057	0.008	-0.083	0.067	2.511	3.143
jvel_LF_thigh	+0.041	0.062	0.107	0.002	-0.077	0.102	3.467	5.308
jvel_LF_calf	+0.037	0.040	0.049	0.001	0.128	0.054	3.860	4.134
jvel_RF_hip	-0.071	-0.218	-0.059	0.005	0.008	-0.061	1.124	3.447
jvel_RF_thigh	-0.028	-0.023	-0.101	0.001	-0.049	-0.099	2.989	2.510
jvel_RF_calf	+0.025	0.052	0.041	0.001	0.117	0.048	3.966	8.155
jvel_LB_hip	+0.070	0.155	0.016	0.005	-0.010	0.015	2.428	5.417
jvel_LB_thigh	-0.011	-0.019	0.032	0.000	-0.032	0.032	2.597	4.455
jvel_LB_calf	+0.046	0.086	-0.121	0.002	0.021	-0.119	2.467	4.610
jvel_RB_hip	+0.053	0.086	-0.107	0.003	0.123	-0.096	2.946	4.758
jvel_RB_thigh	+0.033	0.078	-0.008	0.001	0.043	-0.005	2.437	5.733
jvel_RB_calf	+0.027	0.030	-0.030	0.001	-0.007	-0.030	3.106	3.350
jeff_LF_hip	-0.021	-0.023	0.020	0.000	0.357	0.012	0.836	0.892
jeff_LF_thigh	+0.066	0.072	0.073	0.004	-0.445	0.041	0.867	0.947
jeff_LF_calf	+0.141	0.135	0.019	0.020	-0.096	0.006	0.979	0.939
jeff_RF_hip	+0.030	0.057	0.069	0.001	-0.864	0.020	0.456	0.864
jeff_RF_thigh	-0.014	-0.013	-0.003	0.000	-0.034	-0.002	0.964	0.852
jeff_RF_calf	+0.069	0.071	-0.060	0.005	0.009	-0.060	0.954	0.977
jeff_LB_hip	-0.002	-0.002	-0.062	0.000	0.287	-0.063	0.893	0.944
jeff_LB_thigh	+0.049	0.051	-0.040	0.002	0.241	-0.028	0.894	0.933
jeff_LB_calf	+0.080	0.084	0.018	0.006	0.181	0.033	0.907	0.944
jeff_RB_hip	+0.025	0.026	0.023	0.001	-0.129	0.020	0.908	0.939
jeff_RB_thigh	+0.065	0.077	-0.011	0.004	0.408	0.020	0.803	0.958
jeff_RB_calf	+0.055	0.052	0.001	0.003	0.048	0.003	0.972	0.920
pact_LF_hip	-0.097	-0.081	-0.074	0.009	0.244	-0.094	0.223	0.186
pact_LF_thigh	+0.081	0.048	-0.162	0.007	-0.270	-0.175	0.480	0.286
pact_LF_calf	+0.032	0.037	0.139	0.001	-0.271	0.129	0.188	0.216
pact_RF_hip	-0.041	-0.038	0.506	0.002	-0.613	0.529	0.293	0.273
pact_RF_thigh	+0.120	0.130	0.132	0.014	0.011	0.133	0.261	0.283
pact_RF_calf	-0.073	-0.162	-0.217	0.005	-0.137	-0.195	0.214	0.472
pact_LB_hip	+0.015	0.015	-0.081	0.000	0.184	-0.078	0.269	0.282
pact_LB_thigh	-0.045	-0.080	0.320	0.002	0.377	0.290	0.159	0.284
pact_LB_calf	-0.094	-0.111	0.283	0.009	0.354	0.244	0.262	0.310
pact_RB_hip	-0.001	-0.001	0.355	0.000	-0.068	0.355	0.343	0.264
pact_RB_thigh	+0.048	0.052	0.074	0.002	0.421	0.096	0.219	0.239
pact_RB_calf	+0.036	0.034	-0.114	0.001	0.256	-0.105	0.194	0.189
ACTIONS:								
LF_hip	-0.094	-0.079	0.057	0.009	0.245	0.037	0.222	0.188
LF_thigh	+0.084	0.055	-0.353	0.007	-0.271	-0.368	0.480	0.314
LF_calf	+0.031	0.035	0.010	0.001	-0.271	0.000	0.188	0.215
RF_hip	-0.050	-0.058	-0.065	0.002	-0.613	-0.029	0.293	0.344
RF_thigh	+0.114	0.126	-0.222	0.013	0.011	-0.220	0.261	0.289
RF_calf	-0.072	-0.165	-0.041	0.005	-0.137	-0.019	0.214	0.489
LB_hip	+0.011	0.012	0.002	0.000	0.183	0.004	0.269	0.285
LB_thigh	-0.041	-0.072	0.306	0.002	0.377	0.279	0.159	0.284
LB_calf	-0.091	-0.111	0.107	0.008	0.354	0.068	0.262	0.321
RB_hip	+0.006	0.005	0.178	0.000	-0.067	0.178	0.343	0.269
RB_thigh	+0.063	0.071	0.170	0.004	0.421	0.200	0.219	0.247
RB_calf	+0.034	0.033	0.067	0.001	0.256	0.076	0.194	0.189

Sim → Open-Loop:

Dimension	Corr	Slope	Intcpt	R^2	μ_src	μ_tgt	σ_src	σ_tgt
OBSERVATIONS:								
lin_vel_x	+0.006	0.001	-0.001	0.000	0.056	-0.001	0.118	0.020
lin_vel_y	+0.040	0.008	-0.001	0.002	0.014	-0.001	0.095	0.020
lin_vel_z	-0.068	-0.013	0.000	0.005	0.005	0.000	0.102	0.020
ang_vel_x	-0.053	-0.006	-0.007	0.003	-0.048	-0.007	0.934	0.102
ang_vel_y	+0.020	0.002	-0.004	0.000	-0.150	-0.004	1.129	0.104
ang_vel_z	+0.031	0.005	0.004	0.001	0.048	0.005	0.589	0.102
grav_x	-0.007	-0.000	-0.000	0.000	-0.026	-0.000	0.167	0.010
grav_y	+0.032	0.003	0.000	0.001	-0.010	-0.000	0.097	0.010
grav_z	+0.039	0.003	-0.996	0.002	-0.978	-1.000	0.122	0.010
cmd_x	+0.000	0.000	0.000	0.000	0.070	0.150	0.075	0.000
cmd_y	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
cmd_yaw	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
jpos_LF_hip	+0.126	0.800	0.245	0.016	-0.039	0.214	0.051	0.325
jpos_LF_thigh	+0.007	0.070	-0.111	0.000	0.022	-0.110	0.052	0.499
jpos_LF_calf	+0.070	0.691	-0.413	0.005	0.068	-0.365	0.065	0.645
jpos_RF_hip	+0.118	0.201	-0.472	0.014	0.017	-0.468	0.081	0.139
jpos_RF_thigh	-0.033	-0.182	0.087	0.001	0.051	0.077	0.071	0.393
jpos_RF_calf	+0.081	0.759	-0.135	0.007	-0.040	-0.165	0.077	0.723
jpos_LB_hip	+0.008	0.034	0.151	0.000	0.068	0.154	0.081	0.337
jpos_LB_thigh	+0.017	0.074	0.468	0.000	-0.060	0.463	0.073	0.313
jpos_LB_calf	+0.045	0.152	0.446	0.002	0.071	0.457	0.107	0.363
jpos_RB_hip	+0.136	0.580	-0.094	0.019	0.016	-0.084	0.092	0.392
jpos_RB_thigh	-0.180	-0.579	0.467	0.032	0.038	0.444	0.116	0.374
jpos_RB_calf	-0.042	-0.170	0.237	0.002	-0.041	0.244	0.137	0.552
jvel_LF_hip	-0.031	-0.050	-0.008	0.001	-0.044	-0.006	1.580	2.537
jvel_LF_thigh	-0.018	-0.037	0.037	0.000	0.038	0.035	1.668	3.529
jvel_LF_calf	+0.002	0.004	0.011	0.000	0.064	0.012	1.540	3.814
jvel_RF_hip	-0.009	-0.006	0.003	0.000	-0.001	0.003	1.579	1.030
jvel_RF_thigh	-0.018	-0.027	0.017	0.000	-0.003	0.017	1.976	2.902
jvel_RF_calf	+0.029	0.058	0.115	0.001	-0.022	0.114	2.159	4.263
jvel_LB_hip	-0.016	-0.018	-0.012	0.000	0.065	-0.013	2.153	2.445
jvel_LB_thigh	+0.001	0.001	-0.042	0.000	-0.079	-0.042	2.422	2.441
jvel_LB_calf	+0.059	0.069	-0.033	0.003	0.043	-0.030	2.056	2.417
jvel_RB_hip	+0.037	0.053	-0.005	0.001	-0.095	-0.010	2.091	2.943
jvel_RB_thigh	-0.071	-0.095	-0.040	0.005	0.023	-0.042	1.944	2.583
jvel_RB_calf	+0.003	0.004	0.039	0.000	0.104	0.039	2.305	3.388
jeff_LF_hip	-0.008	-0.013	0.387	0.000	-0.014	0.387	0.525	0.836
jeff_LF_thigh	-0.005	-0.008	-0.432	0.000	-0.012	-0.432	0.525	0.871
jeff_LF_calf	+0.004	0.007	-0.026	0.000	0.013	-0.026	0.531	0.978
jeff_RF_hip	-0.028	-0.028	-0.831	0.001	0.003	-0.831	0.504	0.497
jeff_RF_thigh	+0.010	0.018	-0.032	0.000	-0.016	-0.032	0.559	0.964
jeff_RF_calf	+0.021	0.034	-0.050	0.000	-0.046	-0.052	0.593	0.964
jeff_LB_hip	-0.002	-0.003	0.323	0.000	-0.009	0.323	0.571	0.881
jeff_LB_thigh	+0.032	0.048	0.242	0.001	-0.024	0.240	0.592	0.891
jeff_LB_calf	+0.022	0.038	0.162	0.000	0.038	0.164	0.534	0.919
jeff_RB_hip	-0.043	-0.068	-0.113	0.002	0.010	-0.114	0.574	0.914
jeff_RB_thigh	-0.023	-0.036	0.449	0.001	0.039	0.447	0.504	0.794
jeff_RB_calf	-0.023	-0.038	0.133	0.001	0.023	0.132	0.593	0.956
pact_LF_hip	-0.076	-0.149	0.275	0.006	0.044	0.268	0.124	0.243
pact_LF_thigh	-0.075	-0.148	-0.257	0.006	-0.096	-0.243	0.245	0.486
pact_LF_calf	-0.068	-0.084	-0.271	0.005	-0.005	-0.271	0.145	0.179
pact_RF_hip	-0.036	-0.056	-0.632	0.001	0.041	-0.634	0.189	0.292
pact_RF_thigh	-0.019	-0.026	-0.004	0.000	-0.115	-0.001	0.189	0.259
pact_RF_calf	-0.028	-0.022	-0.162	0.001	-0.101	-0.160	0.271	0.217
pact_LB_hip	+0.049	0.119	0.202	0.002	-0.080	0.193	0.112	0.271
pact_LB_thigh	-0.096	-0.123	0.406	0.009	0.125	0.391	0.126	0.163
pact_LB_calf	+0.096	0.130	0.348	0.009	0.122	0.364	0.194	0.262
pact_RB_hip	+0.156	0.401	-0.142	0.024	0.140	-0.086	0.129	0.333
pact_RB_thigh	+0.064	0.099	0.401	0.004	0.154	0.417	0.149	0.229
pact_RB_calf	-0.015	-0.016	0.272	0.000	0.049	0.271	0.190	0.203
ACTIONS:								
LF_hip	-0.081	-0.154	0.276	0.006	0.046	0.269	0.127	0.243
LF_thigh	-0.073	-0.140	-0.255	0.005	-0.099	-0.242	0.254	0.487
LF_calf	-0.080	-0.098	-0.271	0.006	-0.004	-0.271	0.147	0.179
RF_hip	-0.036	-0.054	-0.632	0.001	0.043	-0.634	0.193	0.292
RF_thigh	-0.032	-0.043	-0.006	0.001	-0.118	-0.001	0.195	0.259
RF_calf	-0.035	-0.027	-0.163	0.001	-0.100	-0.160	0.282	0.217
LB_hip	+0.066	0.154	0.205	0.004	-0.080	0.193	0.116	0.271
LB_thigh	-0.092	-0.117	0.405	0.008	0.125	0.391	0.128	0.163
LB_calf	+0.096	0.126	0.348	0.009	0.121	0.364	0.202	0.263
RB_hip	+0.155	0.398	-0.142	0.024	0.140	-0.086	0.129	0.333
RB_thigh	+0.070	0.105	0.400	0.005	0.157	0.416	0.151	0.229
RB_calf	-0.035	-0.035	0.272	0.001	0.053	0.271	0.202	0.203


LSTM — Observation Correlation: Sim vs Hardware
Pearson Correlation
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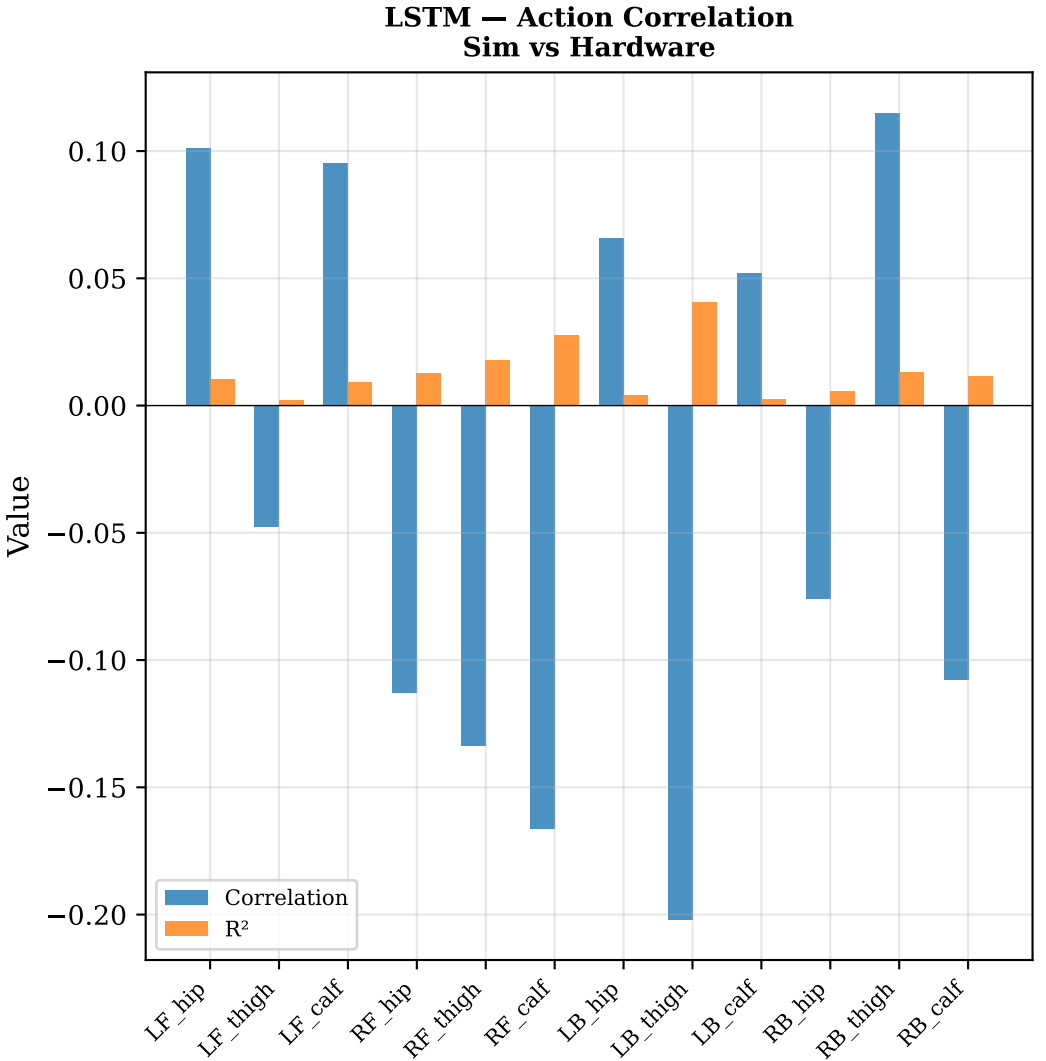
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
lin_vel
ang_vel
gravity
commands
joint_pos
joint_vel
joint_eff
prev_act
lin_vel_x
lin_vel_y
lin_vel_z
ang_vel_x
ang_vel_y
ang_vel_z
grav_x
grav_y
grav_z
cmd_x
cmd_y
cmd_yaw
jpos_LF_hip
jpos_LF_thigh
jpos_LF_calf
jpos_RF_hip
jpos_RF_thigh
jpos_RF_calf
jpos_LB_hip
jpos_LB_thigh
jpos_LB_calf
jpos_RB_hip
jpos_RB_thigh
jpos_RB_calf
jvel_LF_hip
jvel_LF_thigh
jvel_LF_calf
jvel_RF_hip
jvel_RF_thigh
jvel_RF_calf
jvel_LB_hip
jvel_LB_thigh
jvel_LB_calf
jvel_RB_hip
jvel_RB_thigh
jvel_RB_calf
jeff_LF_hip
jeff_LF_thigh
jeff_LF_calf
jeff_RF_hip
jeff_RF_thigh
jeff_RF_calf
jeff_LB_hip
jeff_LB_thigh
jeff_LB_calf
jeff_RB_hip
jeff_RB_thigh
jeff_RB_calf
pact_LF_hip
pact_LF_thigh
pact_LF_calf
pact_RF_hip
pact_RF_thigh
pact_RF_calf
pact_LB_hip
pact_LB_thigh
pact_LB_calf
pact_RB_hip
pact_RB_thigh
pact_RB_calf
R² (linear fit)
1.00
0.75
0.50
0.25
0.00
Observation Dimension


LSTM — Action Correlation
Sim vs Hardware
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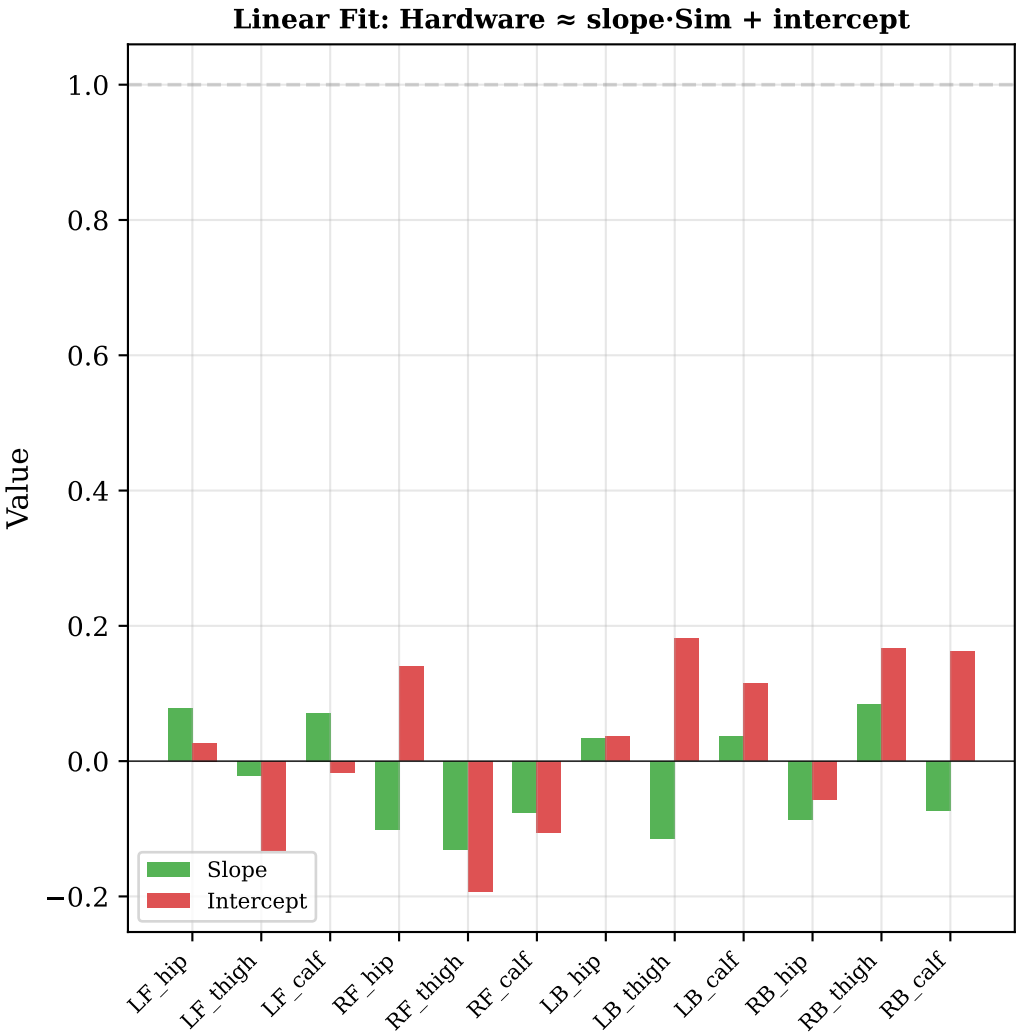
−0.15
−0.20
Correlation
R²
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf


Linear Fit: Hardware ≈ slope·Sim + intercept
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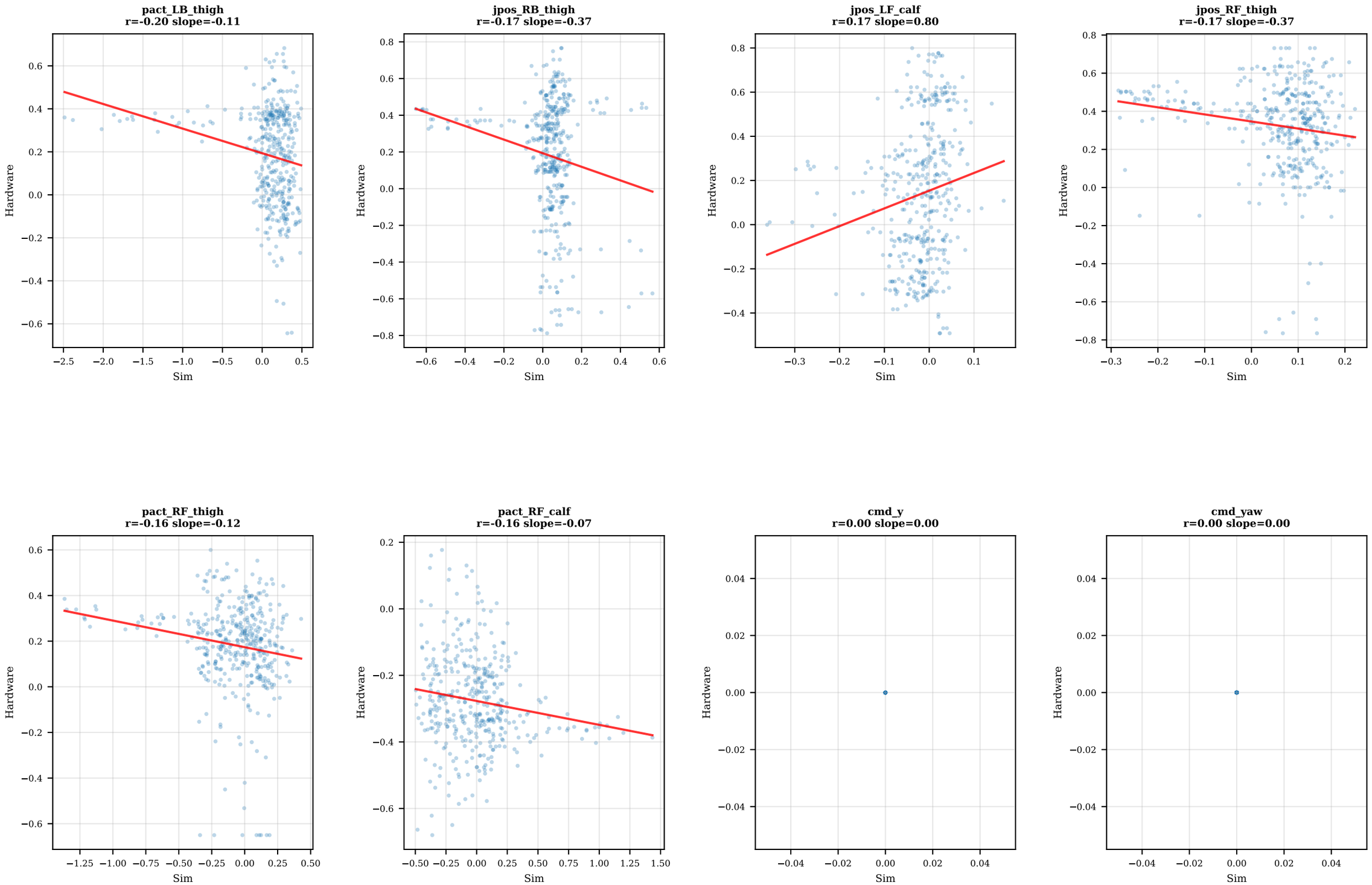
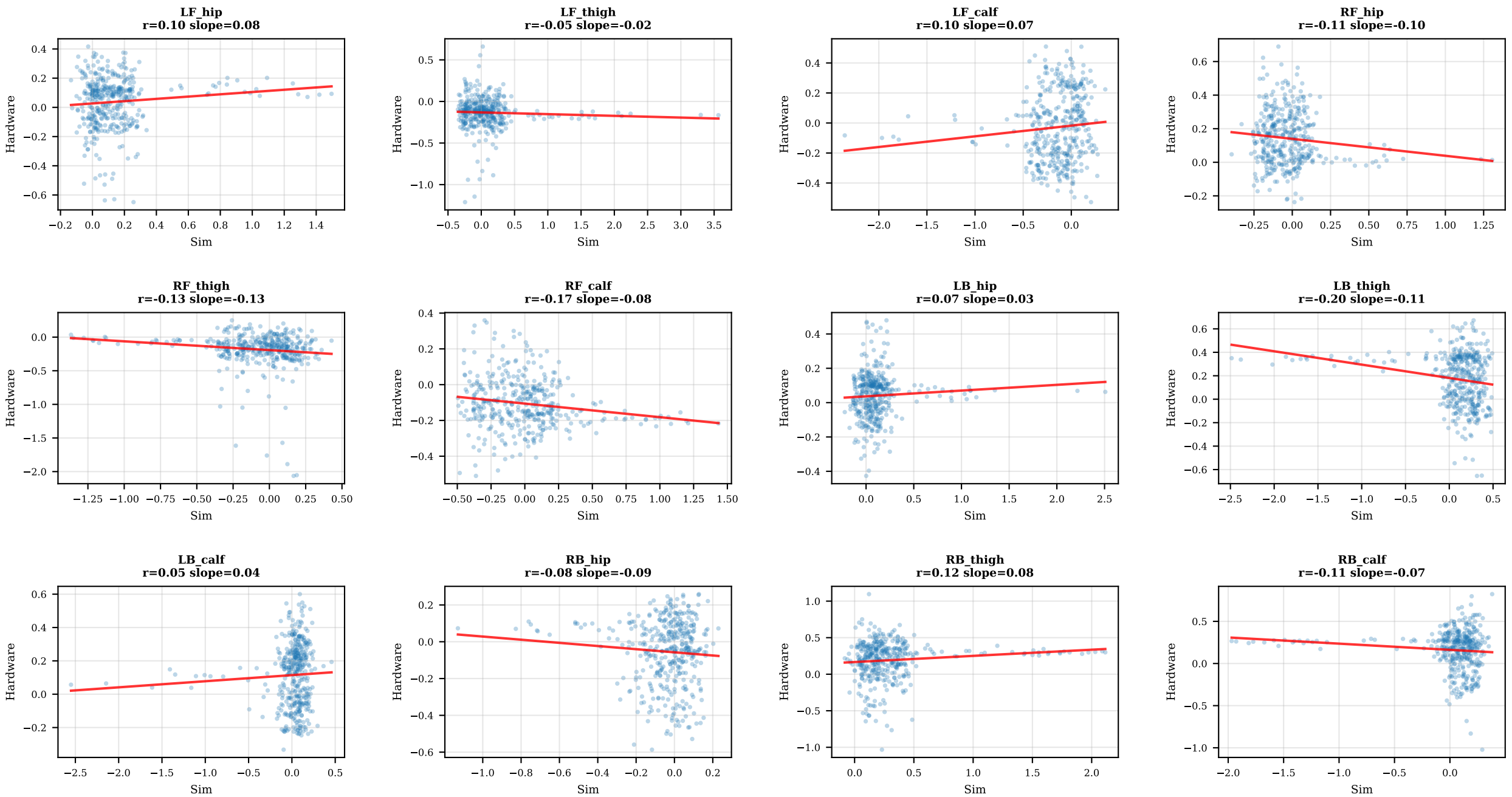
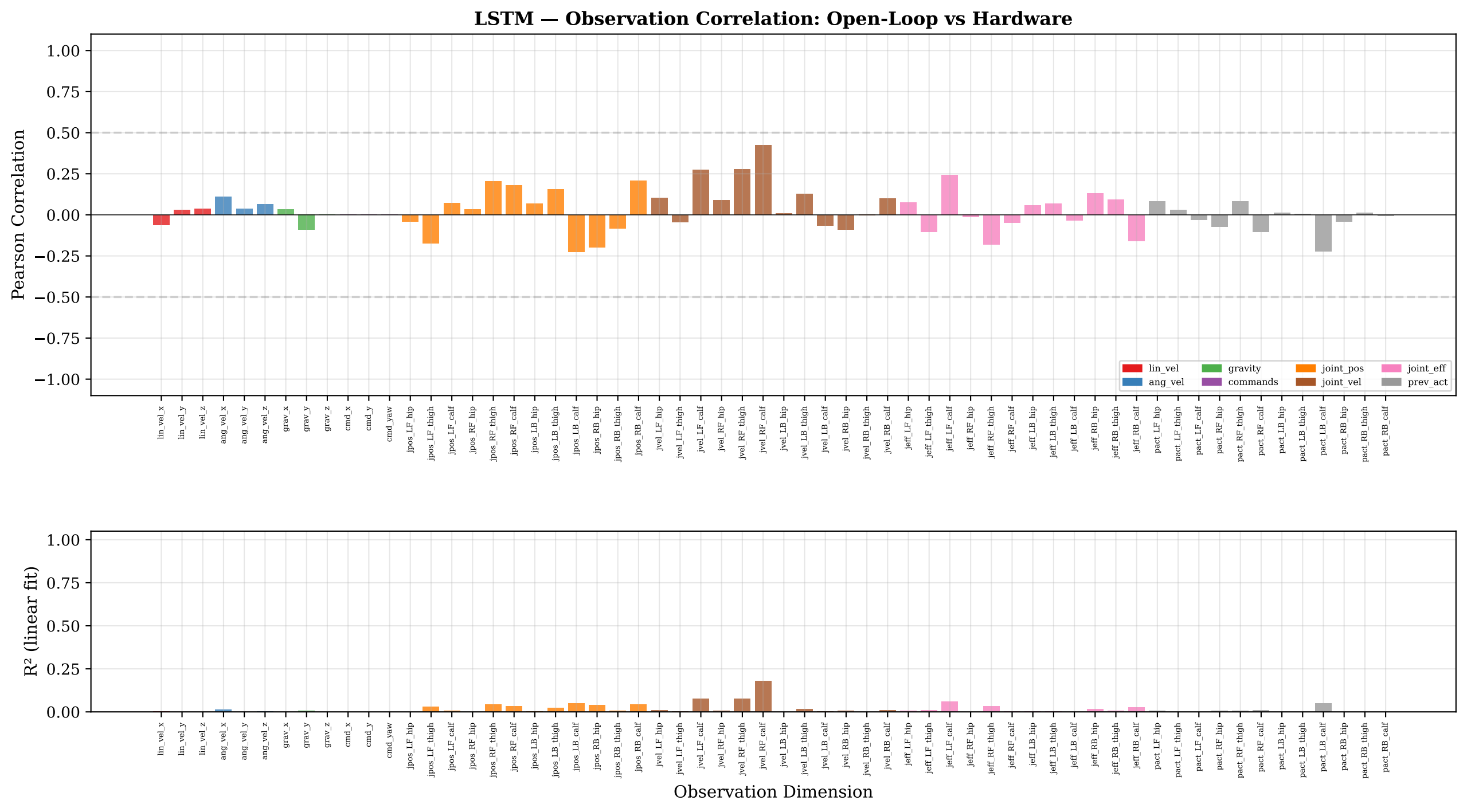
0.4
0.2
0.0
−0.2
Slope
Intercept
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf

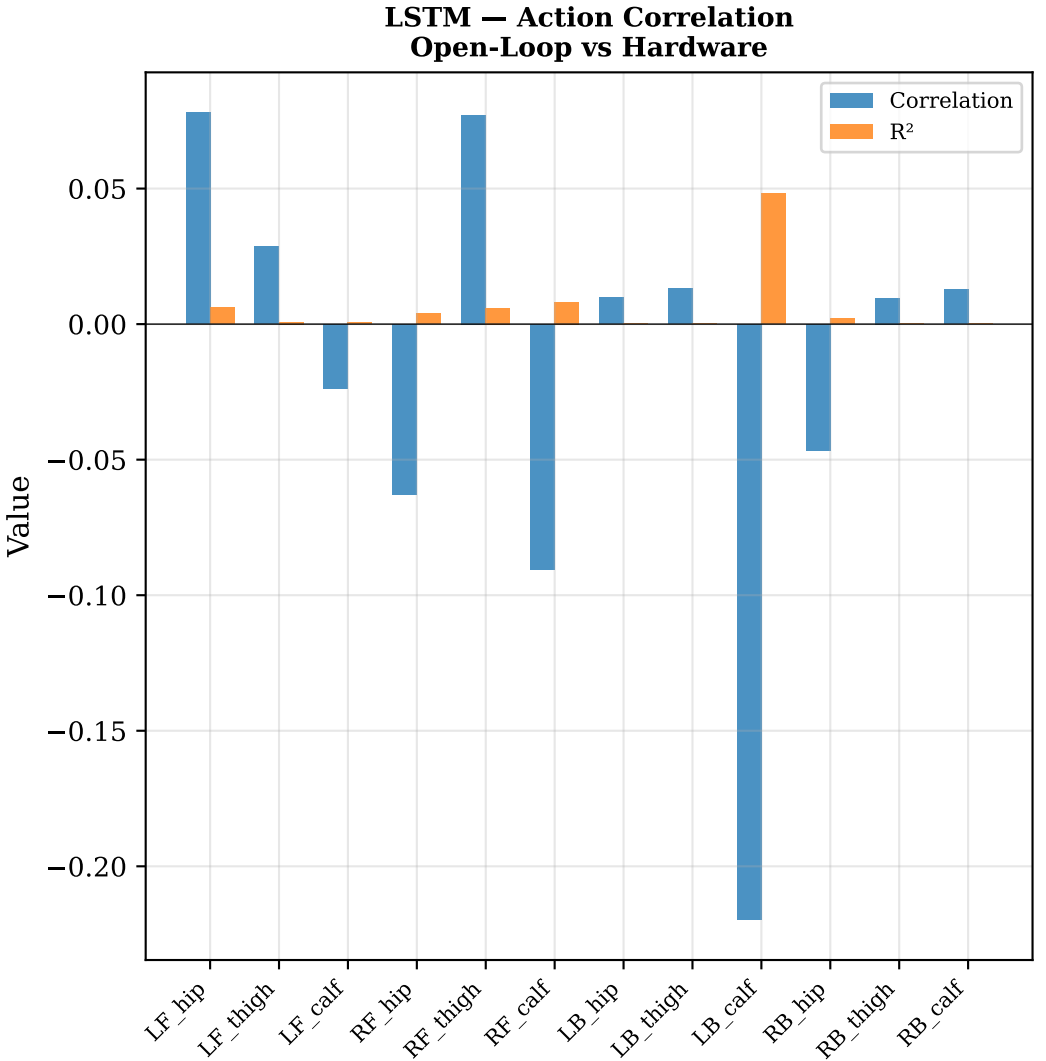
LSTM — Key Dimensions: Sim vs Hardware

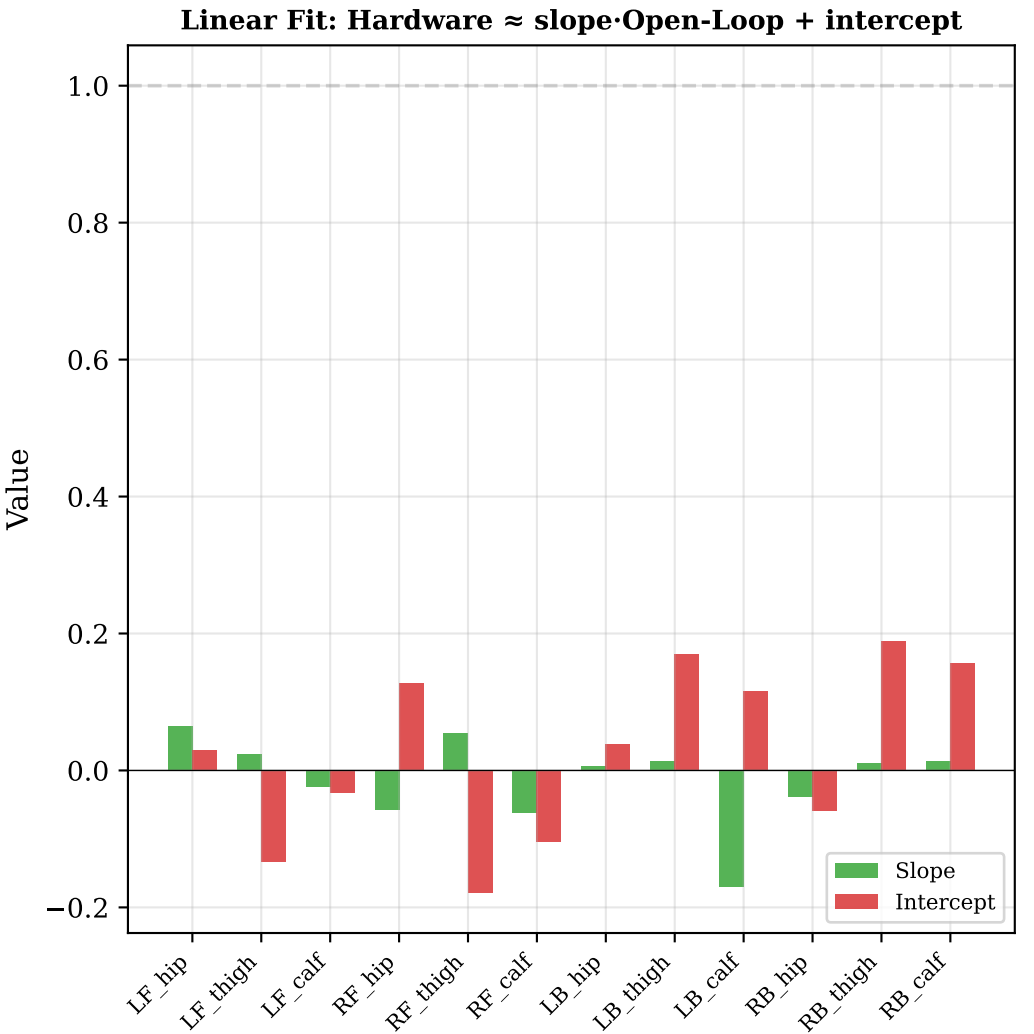


LSTM — Key Dimensions: Sim vs Hardware

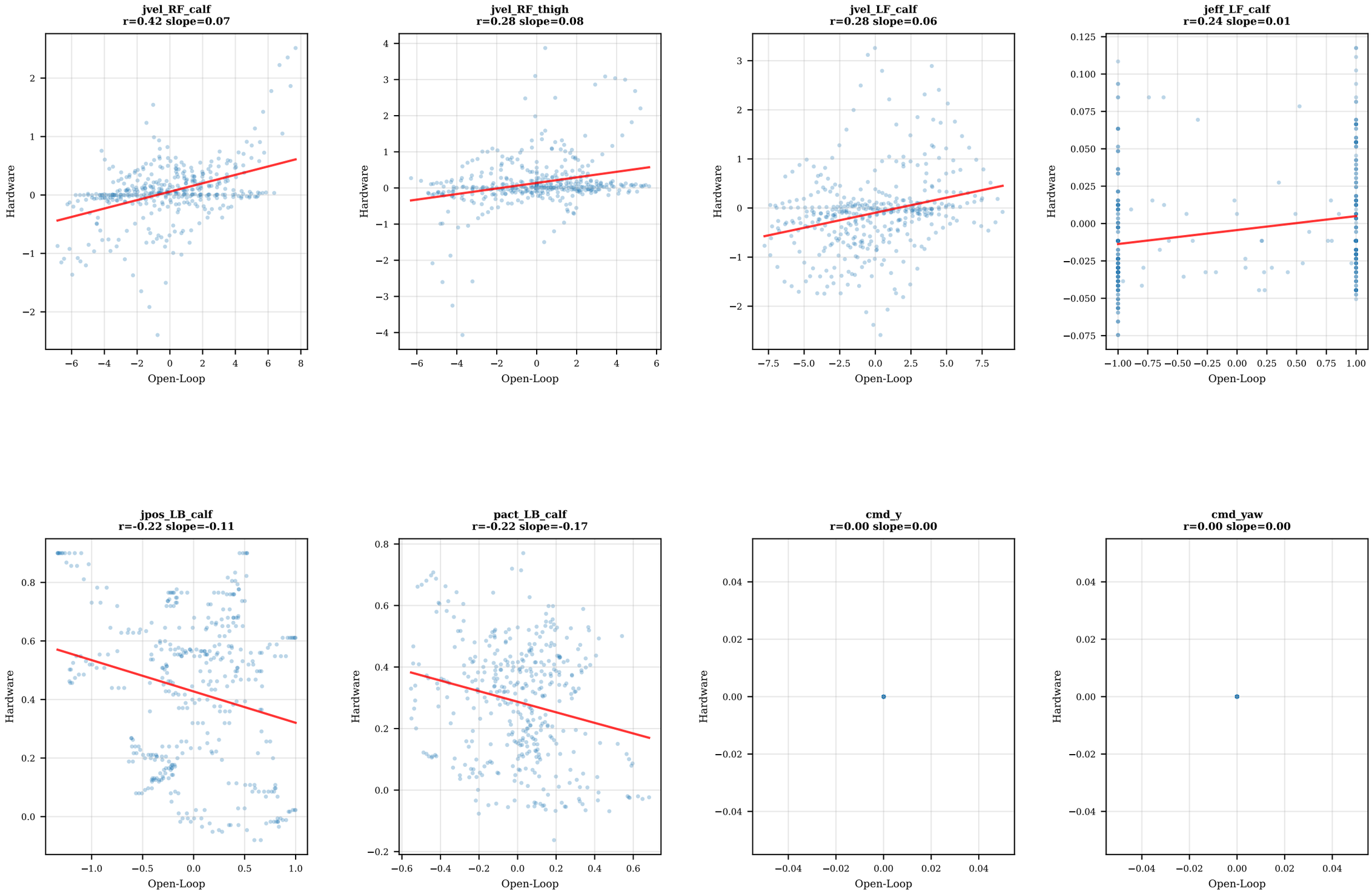



LSTM — Observation Correlation: Open-Loop vs Hardware
Pearson Correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
lin_vel
ang_vel
gravity
commands
joint_pos
joint_vel
joint_eff
prev_act
lin_vel_x
lin_vel_y
lin_vel_z
ang_vel_x
ang_vel_y
ang_vel_z
grav_x
grav_y
grav_z
cmd_x
cmd_y
cmd_yaw
jpos_LF_hip
jpos_LF_thigh
jpos_LF_calf
jpos_RF_hip
jpos_RF_thigh
jpos_RF_calf
jpos_LB_hip
jpos_LB_thigh
jpos_LB_calf
jpos_RB_hip
jpos_RB_thigh
jpos_RB_calf
jvel_LF_hip
jvel_LF_thigh
jvel_LF_calf
jvel_RF_hip
jvel_RF_thigh
jvel_RF_calf
jvel_LB_hip
jvel_LB_thigh
jvel_LB_calf
jvel_RB_hip
jvel_RB_thigh
jvel_RB_calf
jeff_LF_hip
jeff_LF_thigh
jeff_LF_calf
jeff_RF_hip
jeff_RF_thigh
jeff_RF_calf
jeff_LB_hip
jeff_LB_thigh
jeff_LB_calf
jeff_RB_hip
jeff_RB_thigh
jeff_RB_calf
pact_LF_hip
pact_LF_thigh
pact_LF_calf
pact_RF_hip
pact_RF_thigh
pact_RF_calf
pact_LB_hip
pact_LB_thigh
pact_LB_calf
pact_RB_hip
pact_RB_thigh
pact_RB_calf
R² (linear fit)
1.00
0.75
0.50
0.25
0.00
Observation Dimension

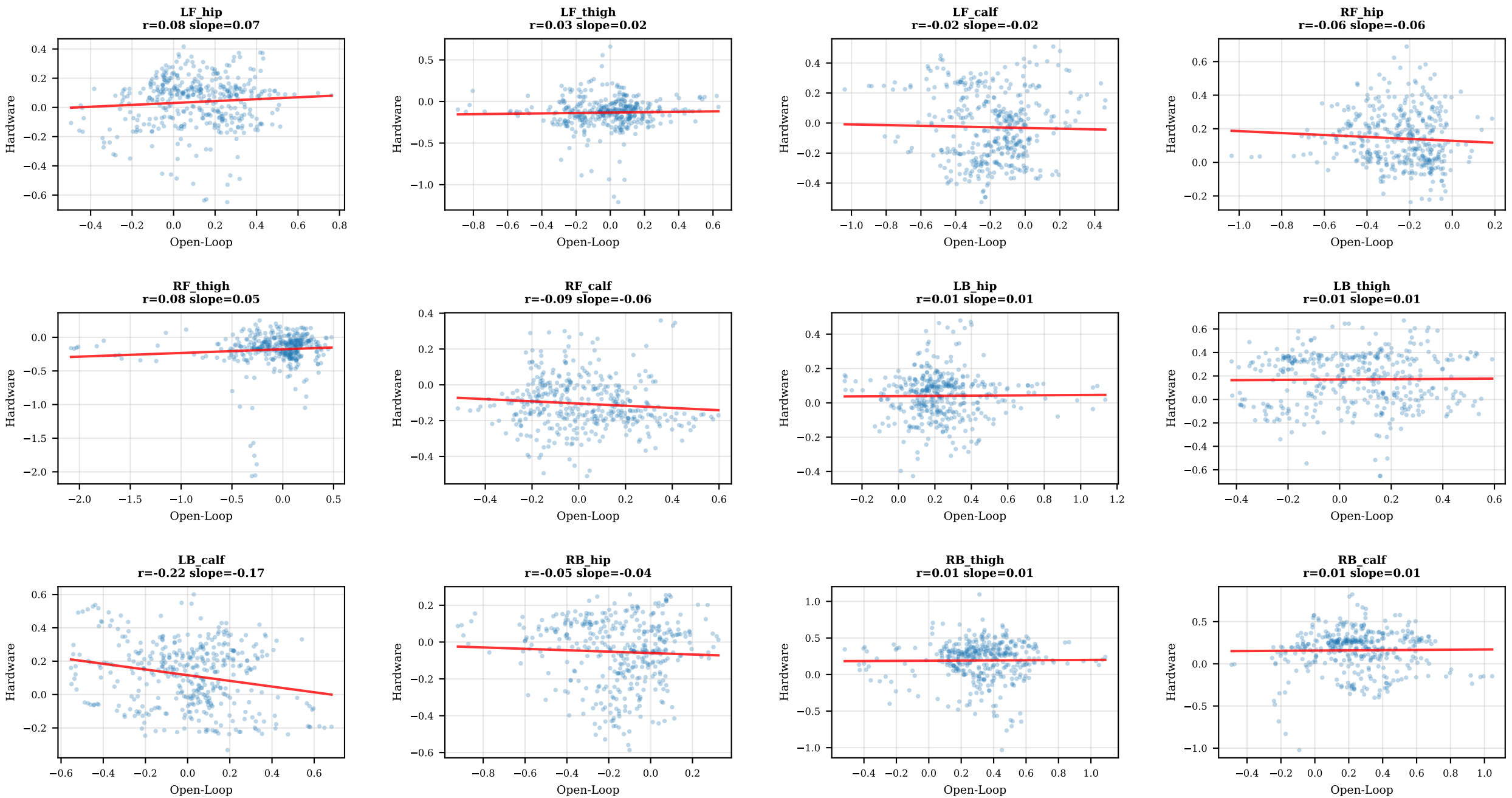

LSTM — Action Correlation
Open-Loop vs Hardware
Correlation
R²
Value
0.05
0.00
−0.05
−0.10
−0.15
−0.20
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf

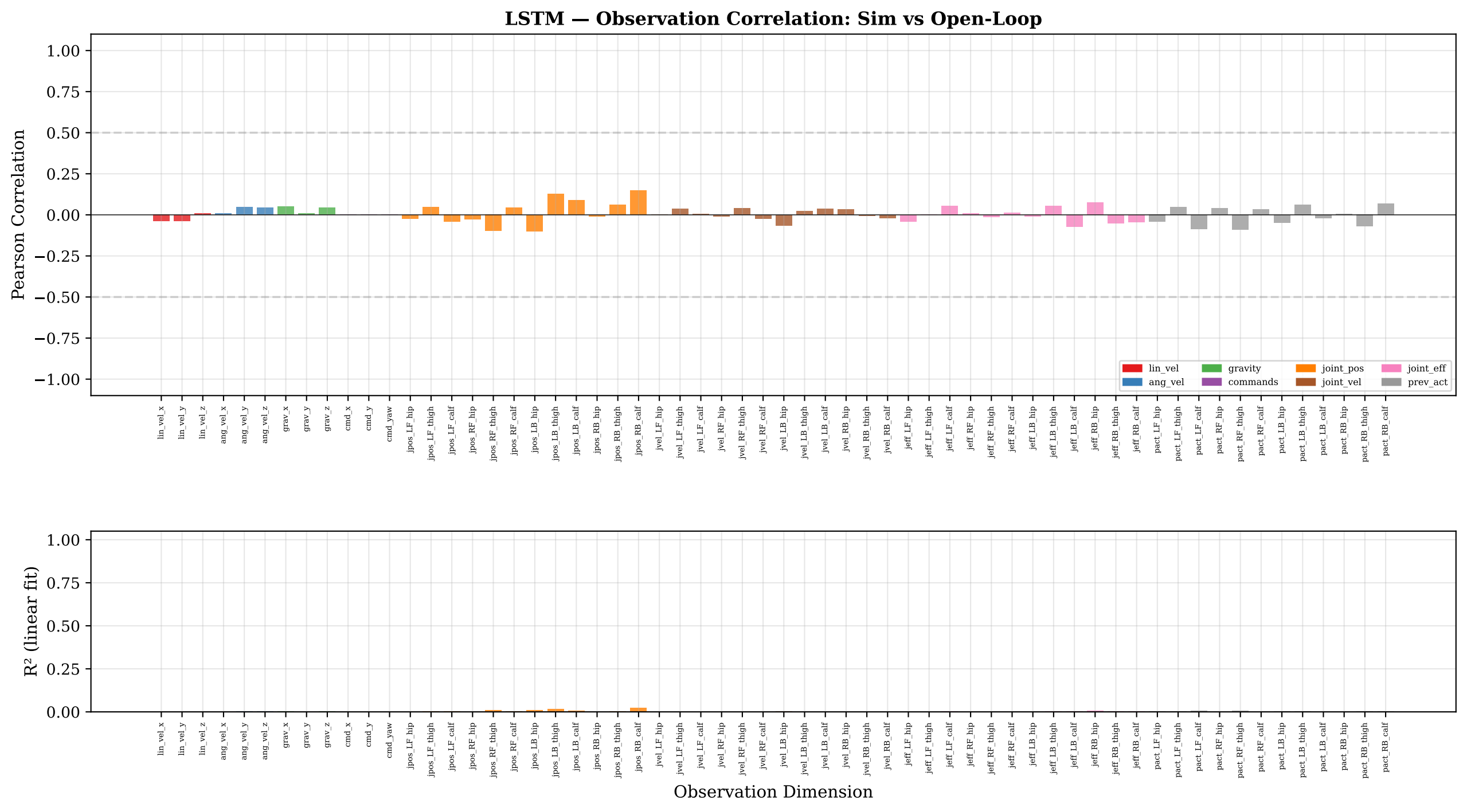

Linear Fit: Hardware ≈ slope·Open-Loop + intercept
Value
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
Slope
Intercept
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf

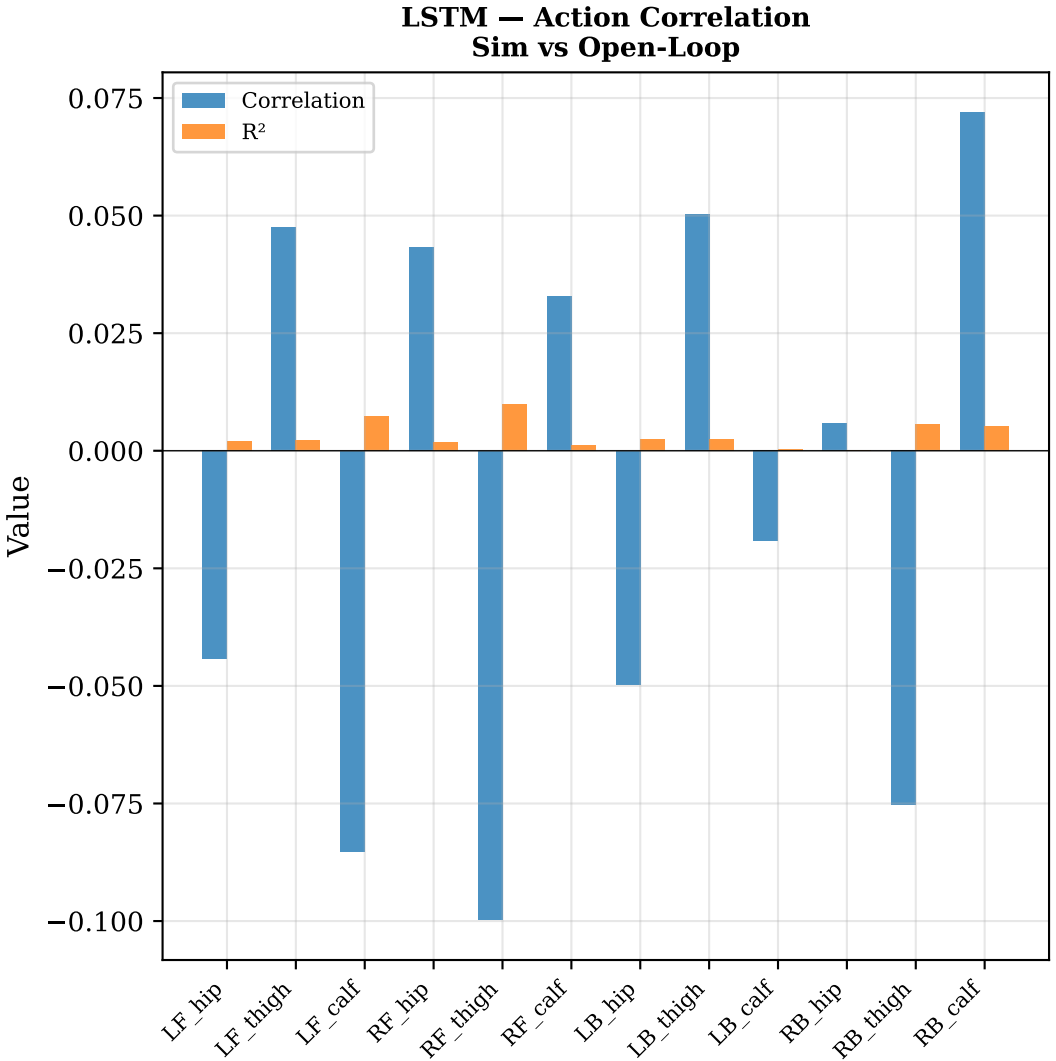
LSTM — Key Dimensions: Open-Loop vs Hardware

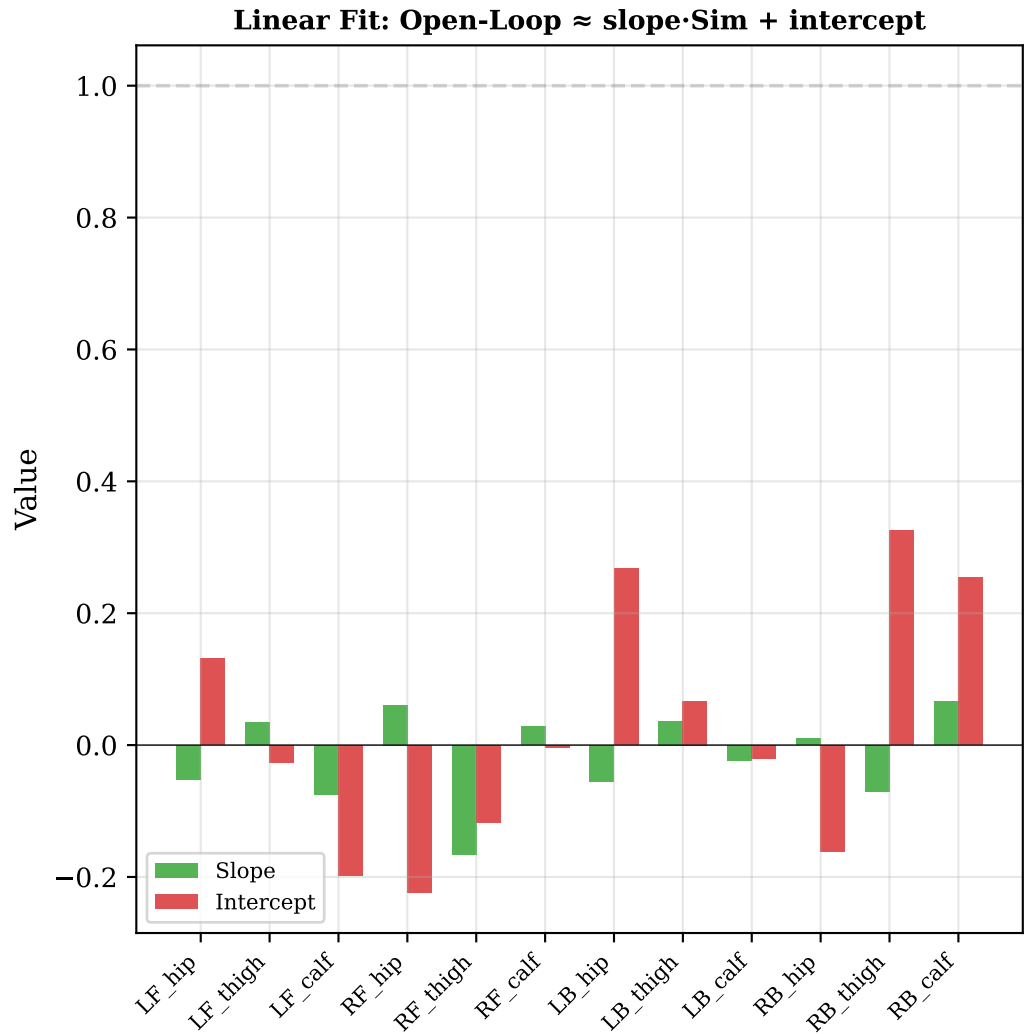


LSTM — Key Dimensions: Open-Loop vs Hardware

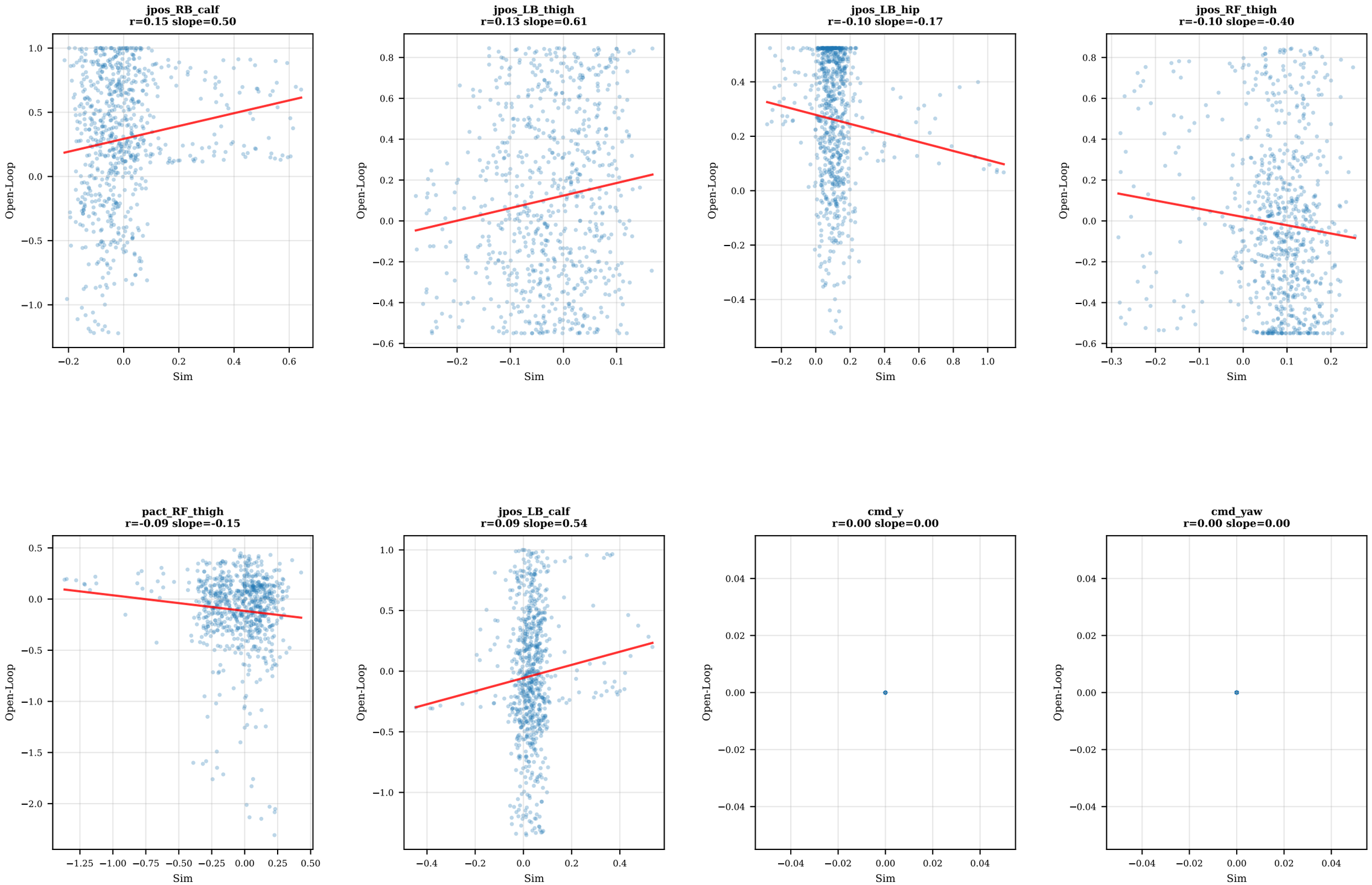



LSTM — Observation Correlation: Sim vs Open-Loop
Pearson Correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
lin_vel
ang_vel
gravity
commands
joint_pos
joint_vel
joint_eff
prev_act
lin_vel_x
lin_vel_y
lin_vel_z
ang_vel_x
ang_vel_y
ang_vel_z
grav_x
grav_y
grav_z
cmd_x
cmd_y
cmd_yaw
jpos_LF_hip
jpos_LF_thigh
jpos_LF_calf
jpos_RF_hip
jpos_RF_thigh
jpos_RF_calf
jpos_LB_hip
jpos_LB_thigh
jpos_LB_calf
jpos_RB_hip
jpos_RB_thigh
jpos_RB_calf
jvel_LF_hip
jvel_LF_thigh
jvel_LF_calf
jvel_RF_hip
jvel_RF_thigh
jvel_RF_calf
jvel_LB_hip
jvel_LB_thigh
jvel_LB_calf
jvel_RB_hip
jvel_RB_thigh
jvel_RB_calf
jeff_LF_hip
jeff_LF_thigh
jeff_LF_calf
jeff_RF_hip
jeff_RF_thigh
jeff_RF_calf
jeff_LB_hip
jeff_LB_thigh
jeff_LB_calf
jeff_RB_hip
jeff_RB_thigh
jeff_RB_calf
pact_LF_hip
pact_LF_thigh
pact_LF_calf
pact_RF_hip
pact_RF_thigh
pact_RF_calf
pact_LB_hip
pact_LB_thigh
pact_LB_calf
pact_RB_hip
pact_RB_thigh
pact_RB_calf
R² (linear fit)
1.00
0.75
0.50
0.25
0.00
Observation Dimension

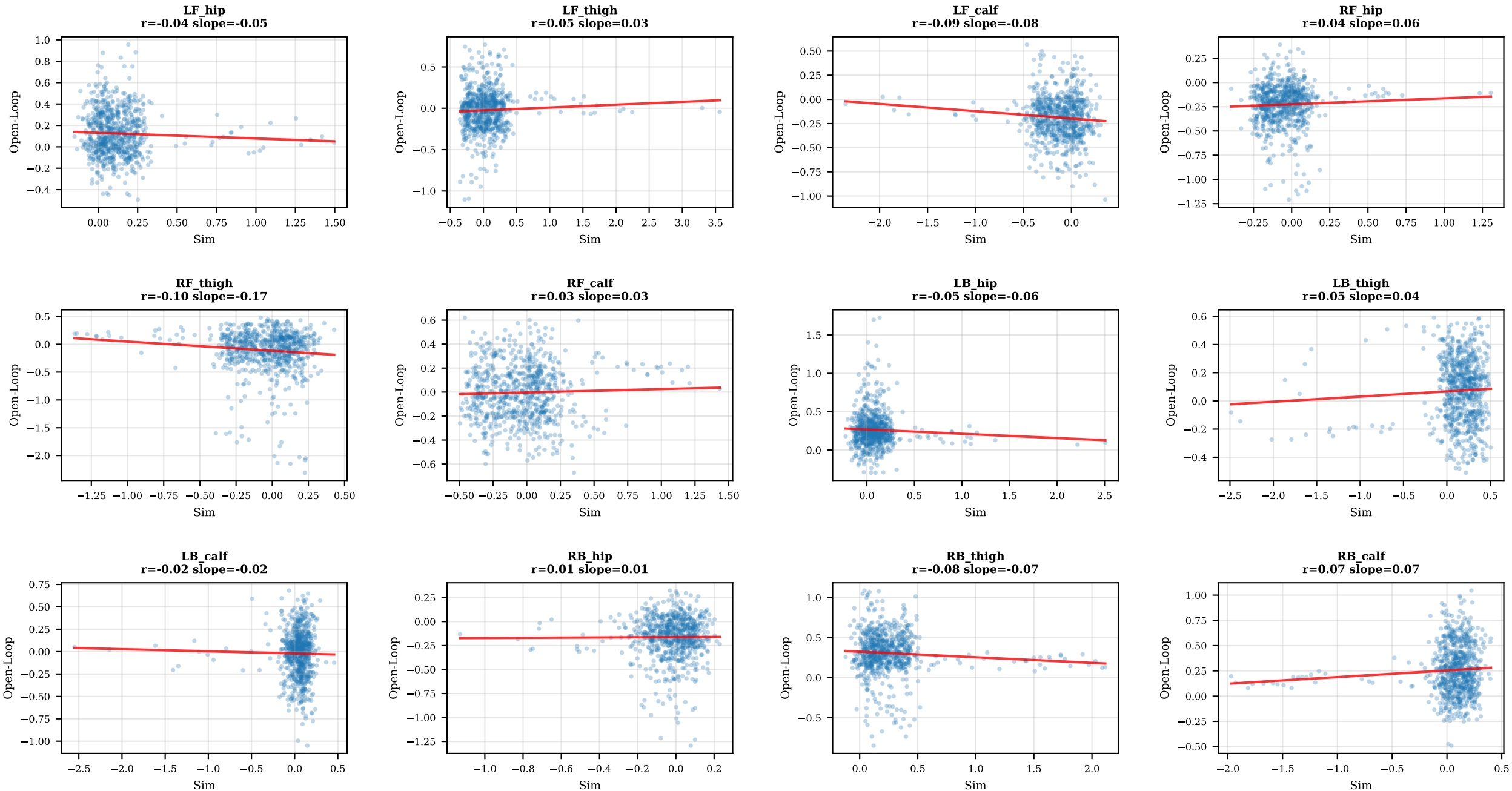

LSTM — Action Correlation
Sim vs Open-Loop
Correlation
R²
Value
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf


Linear Fit: Open-Loop ≈ slope·Sim + intercept
Value
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
Slope
Intercept
LF_hip
LF_thigh
LF_calf
RF_hip
RF_thigh
RF_calf
LB_hip
LB_thigh
LB_calf
RB_hip
RB_thigh
RB_calf

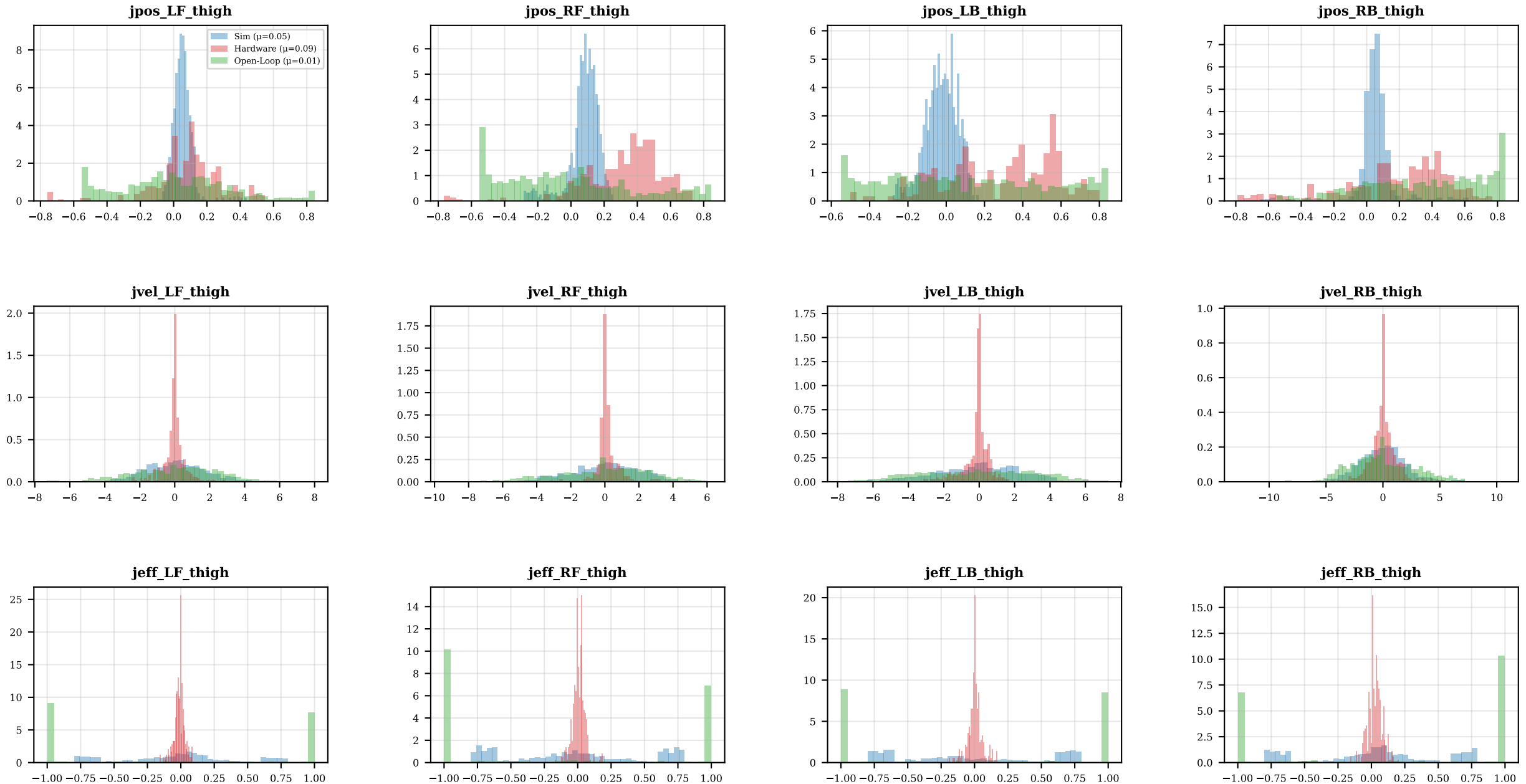
LSTM — Key Dimensions: Sim vs Open-Loop



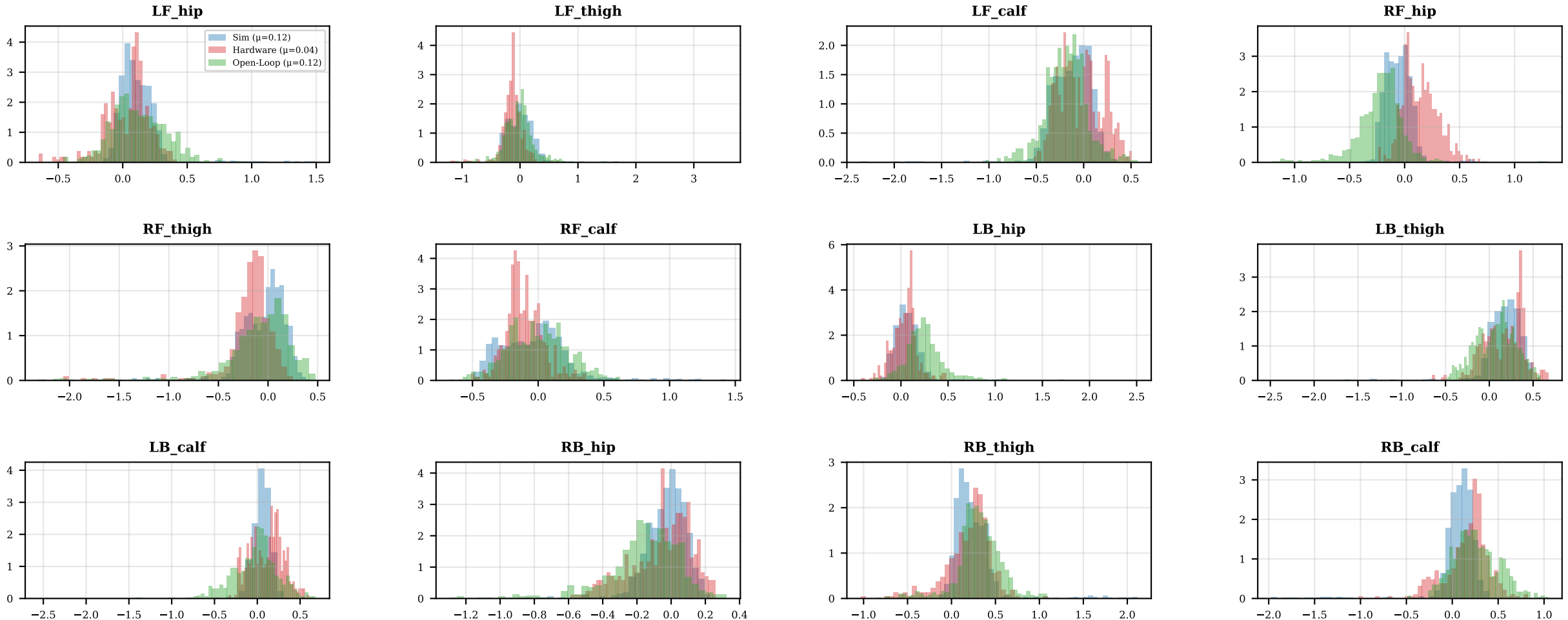
LSTM — Key Dimensions: Sim vs Open-Loop



LSTM — Thigh Obs Distributions Across Domains



LSTM — Action Distributions Across Domains



LSTM — Full Dimension-by-Dimension Linear Relationships

LSTM — LINEAR RELATIONSHIPS: Sim/Open-Loop → Hardware

Sim → Hardware:

Dimension	Corr	Slope	Intcpt	R²	μ_src	μ_tgt	σ_src	σ_tgt
OBSERVATIONS:								
lin_vel_x	-0.012	-0.017	0.101	0.000	0.084	0.100	0.119	0.165
lin_vel_y	-0.076	-0.077	0.028	0.006	0.001	0.028	0.090	0.092
lin_vel_z	-0.004	-0.005	0.180	0.000	0.002	0.180	0.108	0.140
ang_vel_x	-0.001	-0.001	-0.074	0.000	-0.021	-0.074	1.135	0.963
ang_vel_y	+0.003	0.002	-0.088	0.000	-0.249	-0.088	1.369	1.102
ang_vel_z	-0.115	-0.120	-0.065	0.013	-0.041	-0.060	0.684	0.712
grav_x	-0.056	-0.076	-0.301	0.003	-0.051	-0.297	0.196	0.270
grav_y	+0.100	0.135	0.137	0.010	0.010	0.139	0.079	0.106
grav_z	-0.067	-0.068	-0.949	0.005	-0.969	-0.883	0.169	0.171
cmd_x	+0.000	0.000	0.000	0.000	0.150	0.150	0.000	0.000
cmd_y	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
cmd_yaw	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
jpos_LF_hip	+0.107	0.259	-0.123	0.011	0.046	-0.111	0.084	0.203
jpos_LF_thigh	-0.076	-0.203	0.100	0.006	0.057	0.088	0.077	0.208
jpos_LF_calf	+0.172	0.802	0.154	0.029	-0.020	0.138	0.067	0.314
jpos_RF_hip	+0.010	0.003	0.505	0.000	-0.018	0.505	0.070	0.018
jpos_RF_thigh	-0.167	-0.371	0.346	0.028	0.061	0.324	0.106	0.236
jpos_RF_calf	-0.067	-0.062	-0.453	0.004	0.025	-0.455	0.177	0.165
jpos_LB_hip	+0.018	0.016	-0.062	0.000	0.133	-0.060	0.177	0.162
jpos_LB_thigh	-0.105	-0.337	0.273	0.011	-0.033	0.285	0.089	0.286
jpos_LB_calf	+0.120	0.305	0.418	0.014	0.035	0.429	0.103	0.263
jpos_RB_hip	+0.023	0.041	0.197	0.001	-0.047	0.196	0.138	0.250
jpos_RB_thigh	-0.173	-0.371	0.193	0.030	0.038	0.179	0.154	0.329
jpos_RB_calf	+0.130	0.213	-0.063	0.017	0.025	-0.058	0.177	0.291
jvel_LF_hip	+0.009	0.003	-0.133	0.000	0.055	-0.133	1.883	0.629
jvel_LF_thigh	+0.035	0.012	-0.034	0.001	0.138	-0.032	1.749	0.603
jvel_LF_calf	+0.001	0.000	-0.101	0.000	-0.035	-0.101	1.737	0.797
jvel_RF_hip	-0.091	-0.003	-0.010	0.008	0.052	-0.010	1.920	0.056
jvel_RF_thigh	-0.074	-0.028	0.141	0.005	0.059	0.139	1.902	0.716
jvel_RF_calf	-0.046	-0.008	0.050	0.002	0.009	0.050	3.075	0.506
jvel_LB_hip	+0.087	0.018	0.086	0.008	0.110	0.088	2.463	0.508
jvel_LB_thigh	+0.030	0.009	-0.134	0.001	-0.047	-0.135	2.172	0.686
jvel_LB_calf	+0.068	0.024	-0.024	0.005	0.021	-0.023	2.296	0.808
jvel_RB_hip	-0.008	-0.002	-0.084	0.000	-0.035	-0.084	3.145	0.660
jvel_RB_thigh	-0.087	-0.041	0.079	0.008	-0.060	0.081	2.520	1.180
jvel_RB_calf	-0.008	-0.003	0.031	0.000	-0.066	0.032	2.791	0.964
jeff_LF_hip	-0.061	-0.004	-0.008	0.004	-0.009	-0.008	0.460	0.031
jeff_LF_thigh	+0.007	0.001	-0.011	0.000	-0.019	-0.011	0.414	0.039
jeff_LF_calf	+0.016	0.001	-0.004	0.000	-0.011	-0.004	0.419	0.037
jeff_RF_hip	-0.043	-0.001	-0.011	0.002	0.004	-0.011	0.478	0.011
jeff_RF_thigh	-0.060	-0.006	0.008	0.004	-0.057	0.008	0.447	0.046
jeff_RF_calf	+0.024	0.002	-0.008	0.001	-0.022	-0.008	0.598	0.042
jeff_LB_hip	+0.043	0.003	-0.009	0.002	-0.047	-0.009	0.515	0.030
jeff_LB_thigh	+0.003	0.000	0.005	0.000	-0.064	0.005	0.503	0.054
jeff_LB_calf	+0.031	0.002	-0.013	0.001	0.019	-0.013	0.445	0.035
jeff_RB_hip	+0.004	0.000	0.005	0.000	0.046	0.005	0.591	0.039
jeff_RB_thigh	-0.003	-0.000	0.025	0.000	-0.004	0.025	0.492	0.053
jeff_RB_calf	-0.043	-0.004	-0.023	0.002	0.038	-0.023	0.545	0.055
pact_LF_hip	+0.102	0.079	-0.103	0.010	0.135	-0.093	0.222	0.172
pact_LF_thigh	-0.050	-0.021	0.047	0.003	0.084	0.045	0.420	0.175
pact_LF_calf	+0.094	0.071	0.112	0.009	-0.145	0.102	0.297	0.224
pact_RF_hip	-0.105	-0.088	0.705	0.011	-0.030	0.708	0.177	0.149
pact_RF_thigh	-0.160	-0.117	0.174	0.026	-0.071	0.182	0.271	0.198
pact_RF_calf	-0.156	-0.072	-0.277	0.024	-0.006	-0.277	0.302	0.139
pact_LB_hip	+0.064	0.032	-0.043	0.004	0.113	-0.040	0.260	0.132
pact_LB_thigh	-0.201	-0.115	0.194	0.041	0.099	0.182	0.390	0.222
pact_LB_calf	+0.051	0.036	0.286	0.003	0.008	0.286	0.268	0.189
pact_RB_hip	-0.073	-0.082	0.124	0.005	-0.040	0.127	0.156	0.177
pact_RB_thigh	+0.121	0.088	0.064	0.015	0.300	0.091	0.354	0.258
pact_RB_calf	-0.099	-0.067	-0.015	0.010	0.026	-0.016	0.342	0.232
ACTIONS:								
LF_hip	+0.101	0.078	0.027	0.010	0.137	0.038	0.223	0.172
LF_thigh	-0.048	-0.021	-0.132	0.002	0.089	-0.134	0.426	0.184
LF_calf	+0.095	0.071	-0.018	0.009	-0.148	-0.028	0.301	0.224
RF_hip	-0.113	-0.101	0.140	0.013	-0.030	0.143	0.177	0.158
RF_thigh	-0.134	-0.131	-0.193	0.018	-0.075	-0.183	0.275	0.269
RF_calf	-0.166	-0.076	-0.106	0.028	-0.004	-0.106	0.309	0.141
LB_hip	+0.066	0.033	0.037	0.004	0.115	0.041	0.260	0.133
LB_thigh	-0.202	-0.114	0.181	0.041	0.095	0.170	0.399	0.225
LB_calf	+0.052	0.037	0.114	0.003	0.008	0.115	0.268	0.189
RB_hip	-0.076	-0.086	-0.058	0.006	-0.041	-0.054	0.157	0.177
RB_thigh	+0.115	0.084	0.167	0.013	0.306	0.192	0.359	0.263
RB_calf	-0.108	-0.074	0.162	0.012	0.024	0.160	0.351	0.240

Open-Loop → Hardware:

Dimension	Corr	Slope	Intcpt	R²	μ_src	μ_tgt	σ_src	σ_tgt
OBSERVATIONS:								
lin_vel_x	-0.063	-0.492	0.099	0.004	-0.001	0.100	0.021	0.165
lin_vel_y	+0.030	0.134	0.028	0.001	-0.000	0.028	0.020	0.092
lin_vel_z	+0.037	0.260	0.180	0.001	0.000	0.180	0.020	0.140
ang_vel_x	+0.109	1.051	-0.066	0.012	-0.008	-0.074	0.100	0.963
ang_vel_y	+0.038	0.404	-0.084	0.001	-0.011	-0.088	0.103	1.102
ang_vel_z	+0.064	0.464	-0.062	0.004	0.006	-0.060	0.098	0.712
grav_x	+0.033	0.900	-0.296	0.001	-0.000	-0.297	0.010	0.270
grav_y	-0.091	-1.008	0.139	0.008	-0.000	0.139	0.010	0.106
grav_z	-0.000	-0.008	-0.891	0.000	-1.000	-0.883	0.010	0.171
cmd_x	+0.000	0.000	0.000	0.000	0.150	0.150	0.000	0.000
cmd_y	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
cmd_yaw	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
jpos_LF_hip	-0.041	-0.025	-0.108	0.002	0.145	-0.111	0.328	0.203
jpos_LF_thigh	-0.172	-0.108	0.087	0.030	-0.011	0.088	0.332	0.208
jpos_LF_calf	+0.072	0.039	0.148	0.005	-0.267	0.138	0.581	0.314
jpos_RF_hip	+0.032	0.003	0.506	0.001	-0.320	0.505	0.182	0.018
jpos_RF_thigh	+0.206	0.125	0.325	0.042	-0.011	0.324	0.388	0.236
jpos_RF_calf	+0.180	0.058	-0.457	0.032	0.028	-0.455	0.514	0.165
jpos_LB_hip	+0.069	0.046	-0.071	0.005	0.250	-0.060	0.241	0.162
jpos_LB_thigh	+0.154	0.102	0.276	0.024	0.080	0.285	0.429	0.286
jpos_LB_calf	-0.224	-0.107	0.427	0.050	-0.011	0.429	0.551	0.263
jpos_RB_hip	-0.197	-0.157	0.181	0.039	-0.093	0.196	0.315	0.250
jpos_RB_thigh	-0.083	-0.082	0.210	0.007	0.380	0.179	0.334	0.329
jpos_RB_calf	+0.209	0.124	-0.096	0.044	0.303	-0.058	0.492	0.291
jvel_LF_hip	+0.104	0.024	-0.132	0.011	-0.042	-0.133	2.720	0.629
jvel_LF_thigh	-0.045	-0.012	-0.032	0.002	-0.001	-0.032	2.207	0.603
jvel_LF_calf	+0.275	0.061	-0.096	0.076	-0.081	-0.101	3.571	0.797
jvel_RF_hip	+0.088	0.004	-0.010	0.008	-0.046	-0.010	1.364	0.056
jvel_RF_thigh	+0.278	0.077	0.140	0.077	-0.010	0.139	2.585	0.716
jvel_RF_calf	+0.423	0.072	0.057	0.179	-0.096	0.050	2.973	0.506
jvel_LB_hip	+0.008	0.002	0.088	0.000	0.030	0.088	2.191	0.508
jvel_LB_thigh	+0.129	0.027	-0.133	0.017	-0.048	-0.135	3.304	0.686
jvel_LB_calf	-0.066	-0.018	-0.023	0.004	0.044	-0.023	2.914	0.808
jvel_RB_hip	-0.088	-0.023	-0.081	0.008	0.119	-0.084	2.552	0.660
jvel_RB_thigh	-0.004	-0.002	0.081	0.000	-0.141	0.081	2.624	1.180
jvel_RB_calf	+0.100	0.033	0.030	0.010	0.050	0.032	2.917	0.964
jeff_LF_hip	+0.075	0.002	-0.008	0.006	0.090	-0.008	0.941	0.031
jeff_LF_thigh	-0.103	-0.004	-0.011	0.011	-0.084	-0.011	0.942	0.039
jeff_LF_calf	+0.242	0.009	-0.004	0.058	0.001	-0.004	0.968	0.037
jeff_RF_hip	-0.014	-0.000	-0.011	0.000	-0.220	-0.011	0.862	0.011
jeff_RF_thigh	-0.179	-0.009	0.007	0.032	-0.163	0.008	0.924	0.046
jeff_RF_calf	-0.047	-0.002	-0.008	0.002	0.051	-0.008	0.948	0.042
jeff_LB_hip	+0.058	0.002	-0.009	0.003	0.222	-0.009	0.900	0.030
jeff_LB_thigh	+0.069	0.004	0.005	0.005	-0.014	0.005	0.962	0.054
jeff_LB_calf	-0.033	-0.001	-0.013	0.001	-0.004	-0.013	0.940	0.035
jeff_RB_hip	+0.130	0.006	0.007	0.017	-0.252	0.005	0.880	0.039
jeff_RB_thigh	+0.091	0.005	0.024	0.008	0.192	0.025	0.929	0.053
jeff_RB_calf	-0.158	-0.010	-0.022	0.025	0.107	-0.023	0.912	0.055
pact_LF_hip	+0.081	0.068	-0.100	0.007	0.112	-0.093	0.206	0.172
pact_LF_thigh	+0.031	0.025	0.046	0.001	-0.016	0.045	0.219	0.175
pact_LF_calf	-0.029	-0.030	0.096	0.001	-0.189	0.102	0.222	0.224
pact_RF_hip	-0.072	-0.062	0.693	0.005	-0.247	0.708	0.172	0.149
pact_RF_thigh	+0.082	0.043	0.186	0.007	-0.086	0.182	0.380	0.198
pact_RF_calf	-0.103	-0.070	-0.275	0.011	0.017	-0.277	0.206	0.139
pact_LB_hip	+0.013	0.008	-0.042	0.000	0.237	-0.040	0.202	0.132
pact_LB_thigh	+0.006	0.006	0.182	0.000	0.060	0.182	0.225	0.222
pact_LB_calf	-0.221	-0.172	0.287	0.049	0.008	0.286	0.243	0.189
pact_RB_hip	-0.040	-0.033	0.122	0.002	-0.150	0.127	0.216	0.177
pact_RB_thigh	+0.011	0.012	0.087	0.000	0.319	0.091	0.237	0.258
pact_RB_calf	-0.006	-0.006	-0.015	0.000	0.243	-0.016	0.236	0.232
ACTIONS:								
LF_hip	+0.078	0.065	0.030	0.006	0.111	0.038	0.206	0.172
LF_thigh	+0.029	0.024	-0.133	0.001	-0.016	-0.134	0.219	0.184
LF_calf	-0.024	-0.024	-0.033	0.001	-0.188	-0.028	0.223	0.224
RF_hip	-0.063	-0.058	0.128	0.004	-0.247	0.143	0.172	0.158
RF_thigh	+0.077	0.055	-0.178	0.006	-0.086	-0.183	0.380	0.269
RF_calf	-0.090	-0.062	-0.105	0.008	0.017	-0.106	0.207	0.141
LB_hip	+0.010	0.006	0.039	0.000	0.236	0.041	0.202	0.133
LB_thigh	+0.013	0.013	0.169	0.000	0.060	0.170	0.225	0.225
LB_calf	-0.220	-0.171	0.116	0.048	0.008	0.115	0.243	0.189
RB_hip	-0.047	-0.038	-0.060	0.002	-0.150	-0.054	0.217	0.177
RB_thigh	+0.010	0.011	0.189	0.000	0.319	0.192	0.237	0.263
RB_calf	+0.013	0.013	0.157	0.000	0.242	0.160	0.237	0.240

Sim → Open-Loop:

Dimension	Corr	Slope	Intcpt	R²	μ_src	μ_tgt	σ_src	σ_tgt
OBSERVATIONS:								
lin_vel_x	-0.039	-0.008	-0.000	0.001	0.104	-0.001	0.103	0.020
lin_vel_y	-0.036	-0.009	-0.001	0.001	0.007	-0.001	0.085	0.020
lin_vel_z	+0.008	0.002	0.000	0.000	0.001	0.000	0.096	0.020
ang_vel_x	+0.010	0.001	-0.007	0.000	-0.026	-0.007	1.051	0.102
ang_vel_y	+0.047	0.004	-0.004	0.002	-0.123	-0.004	1.169	0.104
ang_vel_z	+0.042	0.007	0.005	0.002	-0.032	0.005	0.616	0.102
grav_x	+0.051	0.003	-0.000	0.003	-0.026	-0.000	0.153	0.010
grav_y	+0.007	0.001	-0.000	0.000	0.015	-0.000	0.073	0.010
grav_z	+0.044	0.003	-0.996	0.002	-0.983	-1.000	0.130	0.010
cmd_x	+0.000	0.000	0.000	0.000	0.150	0.150	0.000	0.000
cmd_y	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
cmd_yaw	+0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
jpos_LF_hip	-0.022	-0.103	0.143	0.001	0.034	0.139	0.069	0.316
jpos_LF_thigh	+0.048	0.251	-0.000	0.002	0.055	0.013	0.064	0.330
jpos_LF_calf	-0.042	-0.441	-0.281	0.002	-0.012	-0.276	0.056	0.594
jpos_RF_hip	-0.026	-0.103	-0.268	0.001	-0.022	-0.266	0.061	0.240
jpos_RF_thigh	-0.098	-0.404	0.019	0.010	0.074	-0.011	0.096	0.394
jpos_RF_calf	+0.042	0.177	-0.020	0.002	0.011	-0.018	0.141	0.592
jpos_LB_hip	-0.099	-0.166	0.279	0.010	0.111	0.260	0.140	0.235
jpos_LB_thigh	+0.128	0.615	0.124	0.016	-0.027	0.107	0.087	0.418
jpos_LB_calf	+0.088	0.541	-0.056	0.008	0.037	-0.036	0.087	0.533
jpos_RB_hip	-0.009	-0.021	-0.099	0.000	-0.042	-0.098	0.123	0.304
jpos_RB_thigh	+0.060	0.167	0.352	0.004	0.050	0.360	0.129	0.359
jpos_RB_calf	+0.150	0.500	0.293	0.022	0.006	0.296	0.153	0.511
jvel_LF_hip	+0.000	0.001	-0.012	0.000	0.072	-0.012	1.773	2.653
jvel_LF_thigh	+0.036	0.049	0.048	0.001	0.061	0.051	1.714	2.352
jvel_LF_calf	+0.004	0.009	-0.136	0.000	0.016	-0.136	1.621	3.657
jvel_RF_hip	-0.010	-0.009	-0.025	0.000	0.018	-0.025	1.806	1.665
jvel_RF_thigh	+0.040	0.051	-0.001	0.002	0.070	0.002	2.031	2.634
jvel_RF_calf	-0.024	-0.029	-0.105	0.001	0.045	-0.106	2.869	3.420
jvel_LB_hip	-0.064	-0.059	0.010	0.004	0.129	0.003	2.323	2.145
jvel_LB_thigh	+0.023	0.030	-0.044	0.001	-0.079	-0.047	2.478	3.182
jvel_LB_calf	+0.037	0.051	-0.014	0.001	-0.026	-0.016	2.053	2.784
jvel_RB_hip	+0.033	0.031	0.009	0.001	-0.051	0.007	2.785	2.569
jvel_RB_thigh	-0.005	-0.005	-0.016	0.000	-0.071	-0.016	2.337	2.718
jvel_RB_calf	-0.019	-0.021	-0.044	0.000	-0.044	-0.043	2.634	2.959
jeff_LF_hip	-0.039	-0.079	0.127	0.002	-0.023	0.129	0.465	0.934
jeff_LF_thigh	+0.003	0.006	-0.077	0.000	-0.023	-0.077	0.430	0.943
jeff_LF_calf	+0.055	0.131	0.012	0.003	0.014	0.014	0.412	0.969
jeff_RF_hip	+0.010	0.017	-0.212	0.000	0.020	-0.212	0.488	0.863
jeff_RF_thigh	-0.012	-0.021	-0.172	0.000	-0.028	-0.171	0.505	0.929
jeff_RF_calf	+0.014	0.021	0.032	0.000	-0.011	0.032	0.608	0.951
jeff_LB_hip	-0.010	-0.017	0.259	0.000	-0.028	0.259	0.515	0.887
jeff_LB_thigh	+0.054	0.089	-0.014	0.003	-0.034	-0.017	0.577	0.954
jeff_LB_calf	-0.070	-0.151	0.011	0.005	0.025	0.007	0.436	0.935
jeff_RB_hip	+0.075	0.115	-0.239	0.006	0.037	-0.235	0.578	0.889
jeff_RB_thigh	-0.051	-0.095	0.175	0.003	0.025	0.172	0.500	0.933
jeff_RB_calf	-0.045	-0.074	0.143	0.002	0.033	0.141	0.555	0.919
pact_LF_hip	-0.042	-0.051	0.131	0.002	0.120	0.125	0.176	0.212
pact_LF_thigh	+0.048	0.035	-0.026	0.002	0.047	-0.025	0.335	0.247
pact_LF_calf	-0.087	-0.078	-0.199	0.008	-0.126	-0.189	0.246	0.221
pact_RF_hip	+0.041	0.058	-0.224	0.002	-0.047	-0.227	0.147	0.208
pact_RF_thigh	-0.091	-0.153	-0.116	0.008	-0.046	-0.109	0.229	0.387
pact_RF_calf	+0.033	0.029	-0.003	0.001	-0.034	-0.004	0.263	0.230
pact_LB_hip	-0.048	-0.054	0.268	0.002	0.081	0.263	0.206	0.231
pact_LB_thigh	+0.062	0.046	0.065	0.004	0.143	0.072	0.306	0.227
pact_LB_calf	-0.018	-0.023	-0.021	0.000	0.031	-0.021	0.205	0.258
pact_RB_hip	+0.006	0.010	-0.162	0.000	-0.025	-0.162	0.132	0.219
pact_RB_thigh	-0.070	-0.066	0.324	0.005	0.249	0.307	0.281	0.264
pact_RB_calf	+0.068	0.064	0.254	0.005	0.059	0.258	0.263	0.246
ACTIONS:								
LF_hip	-0.044	-0.053	0.131	0.002	0.122	0.125	0.177	0.212
LF_thigh	+0.048	0.035	-0.026	0.002	0.050	-0.024	0.339	0.247
LF_calf	-0.085	-0.076	-0.198	0.007	-0.128	-0.188	0.248	0.221
RF_hip	+0.043	0.061	-0.224	0.002	-0.047	-0.227	0.148	0.208
RF_thigh	-0.100	-0.166	-0.117	0.010	-0.048	-0.109	0.233	0.387
RF_calf	+0.033	0.028	-0.004	0.001	-0.033	-0.005	0.267	0.230
LB_hip	-0.050	-0.056	0.268	0.002	0.082	0.263	0.206	0.231
LB_thigh	+0.050	0.037	0.067	0.003	0.141	0.072	0.312	0.227
LB_calf	-0.019	-0.024	-0.021	0.000	0.031	-0.021	0.205	0.258
RB_hip	+0.006	0.010	-0.162	0.000	-0.025	-0.162	0.133	0.219
RB_thigh	-0.075	-0.070	0.325	0.006	0.253	0.307	0.284	0.264
RB_calf	+0.072	0.066	0.254	0.005	0.058	0.258	0.269	0.246